# Multigrid for the even-odd preconditioned Möbius domain-wall operator with a factorized Pauli–Villars coarse correction

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Domain-wall fermions keep chiral symmetry up to effects that vanish as $L_s$ grows, but inverting the Möbius domain-wall operator suffers from critical slowing down toward the physical light-quark mass. Kanamori, Chen and Matsufuru showed that preconditioning with the Pauli–Villars operator and dividing out the site-diagonal part, $(D_{\mathrm{PV}}C_{\mathrm{PV}}^{-1})^{\dagger}DC^{-1}$, moves all eigenvalues into the right half-plane and compresses the high end of the spectrum, so that Krylov methods apply; earlier Pauli–Villars multigrid methods did not outperform red-black CG in four dimensions, as Boyle and Yamaguchi found.

In this work, we construct a two-grid method for its even-odd form in which the quark and Pauli–Villars factors are coarsened separately. Every aggregate contains the whole fifth dimension so that the coarse lattice is four-dimensional with nearest-neighbor couplings, and levels below the coarse lattice replace deflation. These choices address the three reasons Boyle and Yamaguchi gave for why this approach did not beat red-black CG. A single setup serves every value of Neff's $\alpha$. The field of values of the preconditioned operator reaches into the left half-plane, yet its spectrum has $\operatorname{Re}\lambda \geq 0.69$, and its pseudospectra bound the convergence of full GMRES. On four quenched $24^3 \times 48$ configurations the iteration count grows by a factor of 1.6–4.1 from $m = 0.1$ to $m = 0.001$, against 18–27 for mixed-precision even-odd CGNE. At $m = 0.001$ a solve is 2.9–6.0 times faster than mixed-precision CGNE, and a point-source propagator takes half its time, setup included. For future HMC work, which needs solves with $D^{\dagger}$, we derive an exact relation, $D^{\dagger}\Gamma_5 D^{-1}\Gamma_5 = \Gamma_5(1-N)\Gamma_5$ (with $\Gamma_5 = \gamma_5 R$ and $R$ the reflection of the fifth dimension). It makes $\Gamma_5 D^{-1}\Gamma_5$ a preconditioner for these solves without a second multigrid setup.

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## I. Introduction

Many lattice computations need fermions whose chiral symmetry is broken only slightly at finite lattice spacing. Operators of different chirality then do not mix, which weak matrix elements require, as in the Standard Model calculation of $K \to \pi\pi$ decays and $\varepsilon'/\varepsilon$ [1]. Precision calculations at the physical quark masses use such fermions as well: the hadronic vacuum polarization and light-by-light contributions to the muon $g-2$ [2, 3, 4], the $B \to \pi\ell\nu$ form factors on fine lattices with the heavy quark in the same Möbius domain-wall action [5], inclusive semileptonic $B$ decays [6, 7, 8], and parton distribution amplitudes [9]. The overlap operator [10, 11] is exactly chiral on the lattice. Domain-wall fermions [12, 13, 14] cost less: they add a fifth dimension of extent $L_s$, in which a large negative Wilson mass $-M_5$ binds modes of opposite chirality to its two boundaries, and the residual breaking falls as $L_s$ grows. The Möbius generalization [15, 16, 17] adds two coefficients $(b, c)$ that reduce the residual breaking at fixed $L_s$; $c = 0$ is Shamir's operator.

Solving with the domain-wall operator is expensive, because it is neither Hermitian nor positive and one application costs $L_s$ four-dimensional Wilson hops. The standard solver is still the conjugate gradient method [21] on the normal equation of the even-odd operator (CGNE), whose condition number is the square of that of the operator. As the quark mass $m$ decreases, the smallest eigenvalues of $D$ approach $m$, the condition number of the normal equation grows like $1/m^2$ and the CG iterations like $1/m$ until the finite volume limits them: critical slowing down. For Wilson fermions, multigrid removed it. The low modes that slow down Krylov solvers are locally coherent [20]: a few of them, cut into blocks, span the low-mode space. Adaptive aggregation multigrid uses this, and its cost hardly depends on the quark mass [27, 28, 29, 30, 31, 32].

*Multigrid for domain-wall fermions.* For domain-wall fermions multigrid is harder. Because of the large negative Wilson mass, the spectrum encloses the origin. Non-Hermitian Krylov methods then converge only after a number of steps of the order of the system size [35, 36]. Boyle [36] lists four approaches that work to some extent. They differ in the operator that is coarsened.

- Cohen *et al.* [37] coarsened the squared operator, with blocks that cover the whole fifth dimension. Its coarse operator reaches two hops.
- HDCG [38] accelerates CGNE and coarsens the squared even-odd Schur complement. Its coarse stencil has 81 points for blocks of at least $4^4$.
- HDCR [39] coarsens the Hermitian indefinite operator $\gamma_5 R D$ at $c = 0$, where $R$ reflects the fifth dimension. Its coarse operator stays nearest-neighbor, which keeps the setup affordable in HMC.
- Brower, Clark, Howarth and Weinberg [41] noted that the Pauli–Villars preconditioned operator $D_{\mathrm{PV}}^{-1} D$ carries the physical four-dimensional degrees of freedom. They solved $D_{\mathrm{PV}}^{\dagger} D x = D_{\mathrm{PV}}^{\dagger} b$ on all lattice sites. Projecting the Wilson kernel slice by slice gives coarse operators $\hat{D}$ and $\hat{D}_{\mathrm{PV}}$, both nearest-neighbor, and the coarse correction solves $\hat{D}_{\mathrm{PV}}^{\dagger} \hat{D}\, \hat{e} = \hat{r}$. They tested Shamir fermions in the two-dimensional Schwinger model.

Boyle and Yamaguchi [40] compared these approaches for Shamir fermions; on large volumes HDCR was set up and solved faster than a single even-odd CG solve. They also carried the Pauli–Villars method of Ref. [41] to four-dimensional QCD, where it needed about half the fine-matrix applications of red-black CG but 18–31 times its solve time. They gave three reasons. Red-black CG, not the full-lattice solve, is the relevant baseline. The coarse space keeps a fifth dimension of extent $L_s$, so its cost removes the gain. And reducing that cost would need further levels or coarse deflation, which the PV method did not have. They also found that CG converges faster on the Schur complement than on its left-preconditioned form (for the Möbius setups of this work the order is reversed, Appendix D), and that an even-odd smoother did not work with the HDCR correction.

*Recent work by Boyle and collaborators.* The multiple right-hand-side HDCG of Ref. [36] solves the normal equation of the even-odd operator with a two-level CG whose coarse space is spanned by blocked low modes. Treating many right-hand sides at once makes the coarse space cheap, with gains of up to twenty over even-odd CG at physical quark mass. Boyle and Oare [43] coarsen the two-hop operator $D_{\mathrm{PV}}^{\dagger} D$, a 33-point stencil, aiming at solves on a new gauge field as in HMC. Its field of values can reach the left half-plane although its spectrum does not, so the GCR bound of Eisenstat, Elman and Schultz [44] is lost; they add a shift to the smoothers. On $48^3 \times 96$ at the physical quark mass they report 578 fine-matrix applications, against 26500 for red-black CG.

*Setup cost in ensemble generation.* A multigrid setup is paid once per gauge field, which is affordable for propagators but not for HMC. Eick and Mawhinney [52] avoid the setup by blocking a fine Möbius ensemble to a coarse lattice by renormalization-group methods. The interpolated correction then carries high modes that the fine operator amplifies by four orders of magnitude, and they remove them with a Chebyshev filter of degree $150$–$300$.

*The preconditioned operator.* Kanamori, Chen and Matsufuru [45] looked for a preconditioned Möbius operator local enough for a multigrid. Approximating $D_{\mathrm{PV}}$ by its block-diagonal part $B_{\mathrm{PV}}$ brings back eigenvalues with negative real part, so the Pauli–Villars factor has to be kept whole. Dividing out the site-diagonal part $C$ as well gives $A' = (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^{\dagger} D C^{-1}$, whose eigenvalues all lie in the right half-plane, with the smallest condition number among the operators they studied. Neff's form of the domain-wall matrix [46, 47] rescales the fifth dimension by a site-local factor $Q$ and changes the conditioning of the even-odd system.

*This work.* We construct a two-grid method for $A'$ that takes up the Pauli–Villars route of Refs. [41, 40] and addresses the three reasons of Ref. [40] in turn. It is an adaptive aggregation multigrid [27, 30, 32]: the coarse space comes from computed near-null vectors cut into aggregates, and the coarse operators are Galerkin projections. First, the outer system is the even-odd one, as for red-black CG; its inverse contains the inverses of the quark operator and of the adjoint Pauli–Villars operator, and we replace each by a Galerkin inverse. Second, every aggregate contains the whole fifth dimension, so the coarse lattice is four-dimensional and both coarse operators are nearest-neighbor. Third, levels below the coarse lattice with a dense LU at the bottom take the place of deflation for the quark factor, and the Pauli–Villars coarse system is solved by CG on a Tikhonov-regularized normal equation (Sec. III.B). The prolongation lifts each test vector to all lattice sites in two ways, one for each coarse solve, which makes the correction exact on them for the right form (Eq. (3.7)). The smoother is a fixed Chebyshev polynomial, and the setup combines Chebyshev filtering with a bootstrap.

The second result concerns HMC, which solves with $D^{\dagger}$ as often as with $D$. We derive an exact relation, Eq. (4.2), that makes the multigrid of $D$, placed between two $\Gamma_5 = \gamma_5 R$, a preconditioner for $D^{\dagger}$, with no second setup. We test the method on quenched configurations from $12^4$ to $24^3 \times 48$ on a single GPU against even-odd CGNE.

## II. The operator and its preconditioning

### A. Spectra of preconditioned Möbius operators

Where the spectrum lies decides which iterative methods can work. A Krylov method builds a residual polynomial $p$ with $p(0) = 1$ that should be small on the spectrum. Such a polynomial of low degree exists only if a region that excludes the origin, such as an ellipse, encloses the spectrum. The Chebyshev smoother of Sec. III.D, defined by an ellipse in the right half-plane, needs the same. The eigenvalues of the domain-wall operator lie on both sides of the origin, and the preconditioners below move them to one side. Figures 1 and 2, from Ref. [45], show the spectrum without and with both preconditioners; Fig. 9 in Appendix B shows the other preconditioners discussed below.

*The unpreconditioned operator.* The free spectrum (Fig. 1) is a band that reaches into $\mathrm{Re}\,\lambda < 0$. On a $2+1$-flavor configuration at the physical point (Appendix B, Fig. 9) the real parts lie in $[-1.29, 13.2]$. So CG and BiCGStab [22] do not work on $D$ itself, and the practical choice has been CG on a positive definite squared operator.

*Site-diagonal preconditioning.* The site-diagonal part $C$ has an inverse that is the same matrix on every site (Sec. II.B.3). With it, $DC^{-1}$ has real parts in $[-0.22, 2.22]$. The spectrum shrinks, but it still lies on both sides of the origin.

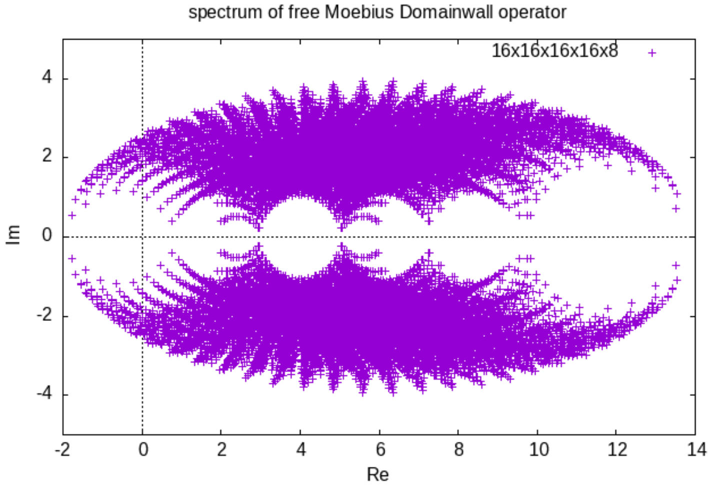

FIG. 1. Spectrum of a free Möbius domain-wall operator on $16^4 \times 8$ with quark mass $0.01$, $b = 1.5$, $c = 0.5$ and domain-wall height $M_5 = 1$. From Ref. [45], Fig. 1.

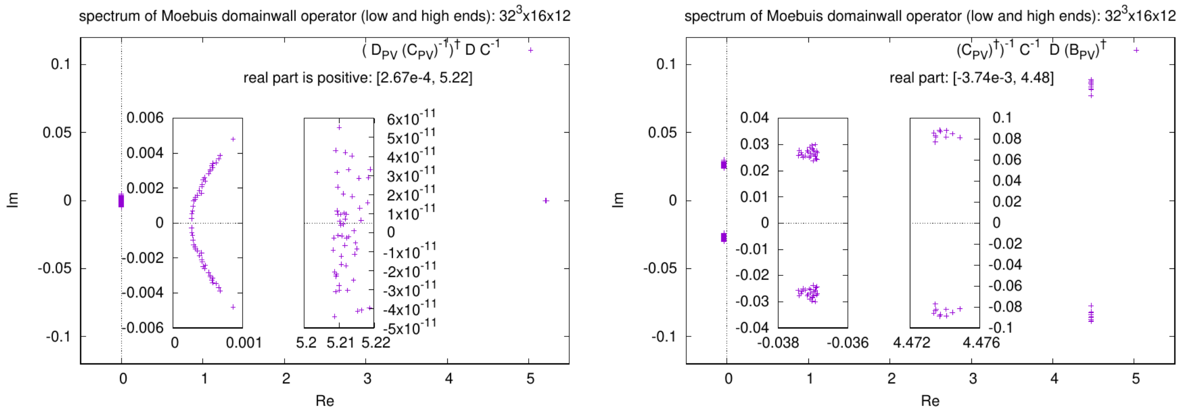

FIG. 2. Left: spectrum of $A' = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger D C^{-1}$, Eq. (2.18), on a JLQCD $2 + 1$-flavor configuration at the physical point; all real parts are positive. Right: the same with $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$; negative real parts reappear. The text of Ref. [45] quotes $48^3 \times 16$ with $L_s = 12$; the panel titles read $32^3 \times 16 \times 12$. From Ref. [45], Fig. 4.

*Pauli–Villars preconditioning.* The operator $D_{\rm PV}^{-1} D$, with $D_{\rm PV} = D(m = 1)$, approximates the overlap operator [41]. Since $D_{\rm PV}^{-1}$ costs as much as a solve with $D$, Ref. [41] used $D_{\rm PV}^\dagger$ instead. $D_{\rm PV}^\dagger D$ has positive real parts. Its low modes lie close to the imaginary axis, shifted by about the quark mass, as for an overlap operator. Its high end, however, reaches about 187, which an overlap operator would not.

*Both together.* The operator $A' = (D_{\rm PV} C_{\rm PV}^{-1})^\dagger D C^{-1}$ has real parts in $[2.67 \times 10^{-4}, 5.22]$ (Fig. 2, left), with the smallest condition number in Ref. [45]. With $D_{\rm PV}$ replaced by its block-diagonal approximation $B_{\rm PV}$, which drops the hops between $4^4$ blocks, negative real parts come back (right). Our method therefore keeps the factor exact on the fine lattice and approximates it only through its Galerkin coarse operator (Sec. III.A).

*Two caveats.* First, the figures show the full-lattice operator $A'$, not the even-odd operators of Sec. II.C. Second, eigenvalues in the right half-plane do not give a field of values there. For a non-normal operator the GCR bound [44] needs the Hermitian part $(A + A^\dagger)/2$ to be positive definite, which can fail while every eigenvalue has positive real part, as Ref. [43] discusses for $D_{\rm PV}^\dagger D$. It fails for the even-odd form and for the preconditioned operator (Sec. V.D.1).

### B. The operator and Pauli–Villars preconditioning

The notation follows Ref. [17] except where stated; Appendix A relates it to Ref. [45] and to Bridge++ symbol by symbol.

#### 1. *The Möbius operator*

*Ingredients.* $\psi$ is a field on the four-dimensional lattice with an extra index $s = 0, \ldots, L_s - 1$, the fifth dimension; $\psi_s$ is an ordinary Dirac field for each $s$. $P_\pm = \frac{1}{2}(1 \pm \gamma_5)$ are the chiral projectors. The Wilson hopping term is

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1-\gamma_\mu)U_\mu(x)\psi(x+\hat{\mu}) + (1+\gamma_\mu)U_\mu^\dagger(x-\hat{\mu})\psi(x-\hat{\mu})\Big], \tag{2.1}$$

and the Wilson operator with mass $-M_5$ (the domain-wall height $M_5 > 0$) is

$$W = (4 - M_5) - \tfrac{1}{2}K. \tag{2.2}$$

This is the usual $D_W(-M_5)$ with hopping coefficient $\frac{1}{2}$; in the $\kappa$ normalization $W = (4 - M_5)(1 - \kappa K)$ with $\kappa = 1/(8 - 2M_5)$. $W$ acts on each slice $s$ in the same way and does not mix slices. Its free spectrum, needed in Sec. IV, is $\{-M_5 + \sum_\mu (1 - \cos p_\mu) \pm i(\sum_\mu \sin^2 p_\mu)^{1/2}\}$.

*The operator.* With

$$D_+ = 1 + bW, \qquad D_- = cW - 1 \tag{2.3}$$

(Ref. [17], Eq. (2.18)), the Möbius operator at quark mass $m$ acts slice by slice as

$$(D\psi)_s = D_+\psi_s + D_-\big(P_-\psi_{s+1} + P_+\psi_{s-1}\big), \qquad \psi_{L_s} \equiv -m\,\psi_0, \quad \psi_{-1} \equiv -m\,\psi_{L_s-1} \tag{2.4}$$

(Ref. [17], Eq. (2.20), with $s$ counted from 0). For $L_s = 4$, as a matrix in $s$,

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{2.5}$$

Shamir's operator is $b = 1$, $c = 0$; the results of this work use $b = 1.5$, $c = 0.5$ ($b - c = 1$, $b + c = 2$). The quark mass enters only through the two corners. The physical four-dimensional quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$.

*The hop in the fifth dimension.* Eq. (2.4) separates into a part that hops in the fifth dimension and one that does not. Define

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(with the same wrap-around)}, \tag{2.6}$$

the off-diagonal part of Eq. (2.5) without $D_-$ and with a minus sign. Then $D = D_+ - D_-Y$. Here $D_-$ stands to the *left* of $Y$, since in Eq. (2.4) it acts on the shifted and projected field, and this order matters in Sec. II.B.2. Inserting Eqs. (2.2) and (2.3),

$$D = 1 + bW - (cW - 1)Y = 1 + Y + W\,(b - cY) = C - \tfrac{1}{2}K\,(b - cY), \qquad C = d_+ + d_-Y, \tag{2.7}$$

with $d_+ = 1 + b(4 - M_5)$ and $d_- = 1 - c(4 - M_5)$. In the last step $W = (4 - M_5) - \frac{1}{2}K$ was split into its site-local constant and its hop; the constant parts form $C$. Note $d_- = -(c(4 - M_5) - 1)$ is *minus* the site-local part of $D_-$, because $Y$ carries the sign. The Pauli–Villars operator is the same operator at unit mass, $D_{\mathrm{PV}} = D|_{m=1}$, $C_{\mathrm{PV}} = C|_{m=1}$; $D_\pm$ and $K$ do not depend on $m$.

### 2. $\Gamma_5$-*transformation and the exact adjoint*

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$, so a solve with $W^\dagger$ is a solve with $W$ between two multiplications by $\gamma_5$. The analogue for domain-wall fermions uses

$$\Gamma_5 = \gamma_5 R, \qquad (R\psi)_s = \psi_{L_s-1-s}. \tag{2.8}$$

Each building block transforms as follows.

1. $W$, $D_+$, $D_-$: they act on every slice alike, so they commute with $R$, and $\gamma_5 W \gamma_5 = W^\dagger$. Hence $\Gamma_5 W \Gamma_5 = W^\dagger$, $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.
2. $Y$: it has no hop in the four dimensions, so $\Gamma_5 Y \Gamma_5$ can be read off Eq. (2.6). $\gamma_5$ commutes with $P_\pm$; $R$ turns the hop $s \to s+1$ into $s \to s-1$. So $\Gamma_5 Y \Gamma_5$ has $-P_-$ on the hop down and $-P_+$ on the hop up, and the corner $-m$ moves to the other corner. This is the transpose of Eq. (2.6) in $s$ with $P_\pm$ in place, which is $Y^\dagger$ since $P_\pm^\dagger = P_\pm$ and $m$ is real: $\Gamma_5 Y \Gamma_5 = Y^\dagger$.
3. $W$ and $Y$ do not commute: $\gamma_\mu$ in $K$ anticommutes with $\gamma_5$, so $\gamma_\mu P_\pm = P_\mp \gamma_\mu$, and the hop of $K$ exchanges the two projectors of $Y$.

From $D = D_+ - D_- Y$ it follows that

$$\Gamma_5 D \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \qquad \text{whereas} \qquad D^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger, \tag{2.9}$$

the adjoint reversing the order of the product. The difference is a commutator,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger = c\,[Y^\dagger, W^\dagger] = \tfrac{c}{2}\big[K^\dagger, Y^\dagger\big], \tag{2.10}$$

where $D_-^\dagger = cW^\dagger - 1$ and $W^\dagger = (4 - M_5) - \frac{1}{2}K^\dagger$ were used. It vanishes for Shamir's operator ($c = 0$) and is of order $c$ otherwise; multigrid on the $\Gamma_5$-Hermitian operator has been used in the $c = 0$ subspace of Möbius fermions (as summarized in Ref. [36]). Thus $\Gamma_5 D \Gamma_5 \neq D^\dagger$ for Möbius fermions: a multigrid built for $D$ cannot be used for $D^\dagger$ by conjugating with $\Gamma_5$. On a $12^4$ configuration with stout links ($M_5 = 1$, $m = 0.001$), $\|\Gamma_5 D \Gamma_5 b - D^\dagger b\|/\|D^\dagger b\| = 0.15$ for $(b, c) = (1.5, 0.5)$, $0.22$ for $(2, 1)$, and $0$ for $(1, 0)$, for a random $b$.

*The exact relation.* It is the generalized $\Gamma_5$-Hermiticity of Ref. [17], Eqs. (2.22)–(2.23), whose $\Gamma_5$ contains $D_-^{-1}$; here $\Gamma_5 = \gamma_5 R$ and $D_-$ appears explicitly. $D_+$ and $D_-$ are functions of the same $W$ and commute. Where $D_-$ is invertible, factor it out to the left:

$$D = D_-\, K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{2.11}$$

$K_D$ is $\Gamma_5$-Hermitian: by item 1, $\Gamma_5 D_-^{-1} D_+ \Gamma_5 = (D_-^\dagger)^{-1} D_+^\dagger$, and this equals $(D_-^{-1} D_+)^\dagger = D_+^\dagger (D_-^\dagger)^{-1}$ because $D_+^\dagger$ and $D_-^\dagger$ commute; with item 2, $\Gamma_5 K_D \Gamma_5 = K_D^\dagger$. Then

$$D^\dagger = K_D^\dagger D_-^\dagger = \Gamma_5 K_D \Gamma_5\, D_-^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger, \tag{2.12}$$

and multiplying by $D_- \Gamma_5$ from the left,

$$\boxed{D_-\, \Gamma_5\, D^\dagger = D\, \Gamma_5\, D_-^\dagger} \tag{2.13}$$

This form has no inverse, holds for any $m$ (so also for $D_{\mathrm{PV}}$, with the same $D_-$), and can be checked by applying both sides to a vector. On the configuration above both sides agree to $3 \times 10^{-16}$ for $(b, c) = (1.5, 0.5)$ and $(2, 1)$. The Hermitian operator of the Möbius case is $\Gamma_5 K_D = \Gamma_5 D_-^{-1} D$; Ref. [17], Eq. (2.22), defines its $\Gamma_5$ as $\gamma_5 R D_-^{-1}$ for that reason. It contains $D_-^{-1}$, whose cost is discussed in Sec. IV.

### *3. The site-diagonal part and its inverse*

$C = d_+ + d_- Y$ of Eq. (2.7) is what remains of $D$ when the four-dimensional hops are dropped. It acts within one four-dimensional site, as an $L_s \times L_s$ matrix in $s$ times the identity in color, and it commutes with $P_\pm$ (only $Y$ is involved, and $Y$ maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$). So it splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. From Eq. (2.6), $Y$ moves $\psi^+$ from $s-1$ to $s$ and $\psi^-$ from $s+1$ to $s$, with the factor $-1$, and the wrap-around gives $+m$:

$$(C\psi)^+_s = \begin{cases} d_+\psi^+_0 + m\, d_-\psi^+_{L_s-1}, & s = 0, \\ d_+\psi^+_s - d_-\psi^+_{s-1}, & s \geq 1, \end{cases} \tag{2.14}$$

a lower bidiagonal matrix with one corner element; for $\psi^-$ the same with $s \to L_s - 1 - s$ (upper bidiagonal). For $L_s = 4$ the block on $\psi^+$ is

$$C^+ = \begin{pmatrix} d_+ & 0 & 0 & m\, d_- \\ -d_- & d_+ & 0 & 0 \\ 0 & -d_- & d_+ & 0 \\ 0 & 0 & -d_- & d_+ \end{pmatrix}. \tag{2.15}$$

*The inverse.* To solve $C^+x = r$, write the rows $s \geq 1$ as $x_s = (r_s + d_- x_{s-1})/d_+$. Starting from an unknown $x_0$, each $x_s$ is affine in $x_0$: with $q = d_-/d_+$,

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \quad p_s = (r_s + d_- p_{s-1})/d_+. \tag{2.16}$$

The row $s = 0$, $d_+ x_0 + m\, d_- x_{L_s-1} = r_0$, with $x_{L_s-1} = p_{L_s-1} + q^{L_s-1}x_0$, fixes $x_0$:

$$x_0 = \frac{r_0 - m\, d_-\, p_{L_s-1}}{d_+ + m\, d_-\, q^{L_s-1}}, \qquad x_s = p_s + q^s x_0. \tag{2.17}$$

For $C^-$ the substitution runs from $s = L_s - 1$ downwards. $C^\dagger$ has the transposed blocks: on $\psi^+$ an upper bidiagonal matrix, solved like $C^-$, and on $\psi^-$ a lower one. $C^{-1}$ is thus exact, costs $O(L_s)$ per site, and does not depend on the gauge field [45]. It exists unless $d_+ + m\, d_- q^{L_s-1} = 0$.

### *4. The preconditioned operator*

Ref. [45] multiplies $D\psi = \eta$ from the left by $(D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}})^\dagger$ and substitutes $\psi = C^{-1}y$:

$$A'y = (D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}})^\dagger\, \eta, \qquad A' = (D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}})^\dagger\, D\, C^{-1}, \qquad \psi = C^{-1}y. \tag{2.18}$$

The two factors are chosen for these reasons:

- For $m = 1$, $DC^{-1} = D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}}$ and $A' = (D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}})^\dagger(D_{\mathrm{PV}}C^{-1}_{\mathrm{PV}})$, which is Hermitian and positive definite. For $m < 1$ the two factors differ only in the corners of Eq. (2.5), and $A'$ is a perturbation of a positive operator.
- $D^{-1}_{\mathrm{PV}}D$ approximates the overlap operator [41]; $D^\dagger_{\mathrm{PV}}$ in place of $D^{-1}_{\mathrm{PV}}$ keeps the cost of one application of $A'$ at two applications of $D$ plus the site-local $C^{-1}$ and $C^{-1}_{\mathrm{PV}}$.

### *5. Link smearing*

The fermion operator may be built from smeared links in place of $U_\mu(x)$; smearing integrates the surrounding gauge field into each link and suppresses the short-distance fluctuations that produce near-zero modes of $W$ [18]. We use the stout projection combined with the APE staples [19]:

$$C_\mu(x) = \sum_{\nu\neq\mu} \rho\Big[U_\nu(x)U_\mu(x+\hat\nu)U^\dagger_\nu(x+\hat\mu) + U^\dagger_\nu(x-\hat\nu)U_\mu(x-\hat\nu)U_\nu(x-\hat\nu+\hat\mu)\Big], \tag{2.19}$$

$$U'_\mu(x) = \exp\Big(\big[C_\mu(x)U^\dagger_\mu(x)\big]_{\mathrm{AT}}\Big)\, U_\mu(x), \tag{2.20}$$

where $[\cdots]_{\mathrm{AT}}$ is the anti-Hermitian traceless part and $\rho$ a tunable parameter. The step (2.20) is applied repeatedly; "stout $\rho \times n$" means $n$ steps with the same $\rho$. Only $W$ (through $K$) sees the smeared links; $Y$, $C$ and $Q$ do not depend on the gauge field.

#### 6. *Neff's* $\alpha$

Neff's form of the domain-wall matrix [46, 47] is $D^{(\alpha)} = DQ$ with a site-local $Q$, diagonal in the fifth dimension,

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \ \ (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{2.21}$$

The following properties, each read off Eq. (2.21), are used below:

- $Q$ is Hermitian, and $Q^{-1}$ is Eq. (2.21) with $\alpha \to 1/\alpha$.
- $Q$ leaves the components $P_-\psi_0$ and $P_+\psi_{L_s-1}$ unchanged, which make up the quark field $q$; so $D^{(\alpha)}$ and $D$ have the same four-dimensional propagator.
- The solution of $D\psi = \eta$ is $\psi = Q\psi^{(\alpha)}$ with $D^{(\alpha)}\psi^{(\alpha)} = \eta$, since $D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} = D\psi$.
- $Q$ is site-local, so the site-diagonal part of $DQ$ is $CQ$, with $(CQ)^{-1} = Q^{-1}C^{-1}$ and $(CQ)^\dagger = QC^\dagger$.

The factor may be applied to the quark operator alone or to the Pauli–Villars operator as well; we write $D^{(\alpha)}_{\text{PV}} = D_{\text{PV}}Q$ in the latter case. For $\alpha < 1$ it improves the conditioning of the even-odd operator [46, 47].

### C. Even-odd forms

*The diagonal blocks are* $C$. Order the sites so that the even ones come first. We write $X_{pq}$ for the block of an operator $X$ that maps the sites of parity $q$ to those of parity $p$, and $\psi_e$, $\psi_o$ for the even and odd parts of a field. The hopping term $K$ connects nearest neighbors, which always have opposite parity, so it has no even–even or odd–odd block. In $D = C - \frac{1}{2}K(b - cY)$ (Eq. (2.7)) the diagonal blocks therefore come from $C$ alone and the off-diagonal ones from the hop alone:

$$D = \begin{pmatrix} D_{ee} & D_{eo} \\ D_{oe} & D_{oo} \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b - cY) \ \ (p \neq q), \qquad C = \begin{pmatrix} C_e & 0 \\ 0 & C_o \end{pmatrix}. \tag{2.22}$$

So the $D_{ee}$ and $D_{oo}$ of the even-odd literature (Ref. [47], Eq. (18); Ref. [45], Eq. (4)) *are* the site-diagonal part: $D_{ee} = C_e$ and $D_{oo} = C_o$ are the blocks of the same $C$ that appears in $A' = (D_{\text{PV}}C_{\text{PV}}^{-1})^\dagger DC^{-1}$, restricted to the even and to the odd sites. Since $C$ is the same $L_s \times L_s$ matrix on every site (Sec. II.B.3), $C_e$ and $C_o$ are also the same matrix, applied on different sites. The same holds for $D_{\text{PV}}$ with $C_{\text{PV},e}$, $C_{\text{PV},o}$.

With these blocks the Schur complement on the even sites is $S = C_e - D_{eo}C_o^{-1}D_{oe}$, and $S_{\text{PV}}$ is that of $D_{\text{PV}}$. The injection $\iota$ puts a field on the even sites onto the whole lattice, with zero on the odd sites, and the projection $\pi = \iota^\dagger$ takes the even sites:

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e, \qquad \iota\pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \begin{pmatrix} \psi_e \\ 0 \end{pmatrix}, \tag{2.23}$$

Two identities are used throughout, $S^{-1} = \pi D^{-1}\iota$ and $(S_{\text{PV}}^\dagger)^{-1} = \pi(D_{\text{PV}}^\dagger)^{-1}\iota$. To see them, solve $D\psi = \iota\eta_e = (\eta_e, 0)^\mathsf{T}$. The odd rows give $\psi_o = -C_o^{-1}D_{oe}\psi_e$, and the even rows then read $S\psi_e = \eta_e$. Hence $S^{-1}\eta_e = \pi D^{-1}\iota\eta_e$, and the same argument for $D_{\text{PV}}^\dagger$ gives the second identity.

*Two forms.* The site-diagonal factor can stand on either side of $S$. This gives the right and the left form,

$$\tilde{A}_R = (S_{\text{PV}}C_{\text{PV},e}^{-1})^\dagger \, (S\, C_e^{-1}), \qquad \tilde{A}_L = (C_{\text{PV},e}^{-1}S_{\text{PV}})^\dagger \, (C_e^{-1}S). \tag{2.24}$$

$\tilde{A}_R$ is the even-odd counterpart of $A'$. $C_e^{-1}S$ is the usual left-preconditioned even-odd operator, which Boyle and Yamaguchi found worse than $S$ itself for CG on the normal equation [40].

*With Neff's $\alpha$.* Replace $D$ by $DQ$. Since $Q$ acts within a site and is diagonal in $s$, the blocks of $DQ$ are $C_eQ_e$, $D_{eo}Q_o$, $D_{oe}Q_e$ and $C_oQ_o$. Its Schur complement is therefore

$$C_eQ_e - D_{eo}Q_o(C_oQ_o)^{-1}D_{oe}Q_e = SQ_e, \tag{2.25}$$

and its diagonal block is $C_eQ_e$. In the right form the two factors of $Q_e$ cancel: $SQ_e(C_eQ_e)^{-1} = SC_e^{-1}$. So $\tilde{A}_R$ does not depend on $\alpha$. In the left form they do not cancel. With $(C_eQ_e)^{-1}SQ_e = Q_e^{-1}C_e^{-1}SQ_e$ and $Q$ Hermitian,

$$\tilde{A}_L^{(\alpha)} = \begin{cases} Q_e\,(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-2}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ and } D_{\mathrm{PV}}, \\ (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\,Q_e^{-1}\,(C_e^{-1}S)\,Q_e, & \alpha \text{ in } D \text{ only.} \end{cases} \tag{2.26}$$

For $D\psi = \eta$, with $\eta_e' = \eta_e - D_{eo}C_o^{-1}\eta_o$, the right-hand side is $\rho_R = (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger\eta_e'$ or $\rho_L = (C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger C_e^{-1}\eta_e'$ (operators with $Q$ when $\alpha \neq 1$); the solution is $\psi_e^{(\alpha)} = C_e^{-1}y$ (right) or $y$ (left), $\psi_o^{(\alpha)} = C_o^{-1}(\eta_o - D_{oe}\psi_e^{(\alpha)})$, $\psi = Q\psi^{(\alpha)}$.

*Why both forms.* The two forms serve different purposes. The factorized correction of Sec. III.A is exact for the right form (Eq. (3.7)), but $Q$ cancels in it, so the right form cannot use Neff's $\alpha$. The left form keeps $\alpha$. The correction is not exact for it, yet it converges within a few iterations of the right form, and with $\alpha$ it can save iterations. The outer solver therefore uses the left form in practice. The measurements of Sec. V are on the left form unless stated.

*How $\alpha$ acts on CG and on the multigrid.* The left Schur factor of $DQ$ is a similarity transform of that of $D$:

$$\mathcal{K}^{(\alpha)} = (C_eQ_e)^{-1}SQ_e = Q_e^{-1}\mathcal{K}Q_e, \qquad \mathcal{K} = C_e^{-1}S. \tag{2.27}$$

So $\mathcal{K}^{(\alpha)}$ and $\mathcal{K}$ have the same eigenvalues but different singular values, because $Q_e$ is not unitary. CG on the normal equation converges at a rate set by the singular values and sees the change at every mass. In the multigrid the coarse correction removes the low modes for any $\alpha$, so we expect $\alpha$ to help it less than CG. At $m = 0.001$ on $24^3 \times 48$, $\alpha = 0.6$ saves CG 25–35% of its iterations and the multigrid at most three (Sec. V.B.1).

## III. The two-grid method

A multigrid preconditioner removes the error of an iterate in two parts. A few steps of a cheap iteration on the fine lattice, the *smoother*, damp the high modes. The low modes, which make CG slow, are removed by solving a smaller problem on a coarse lattice. The fine lattice is cut into blocks, here of $4^4$ sites with the whole fifth dimension; each block becomes one coarse site, whose degrees of freedom are a few approximate low modes restricted to the block, the *test vectors*. The prolongation $\mathcal{P}$ maps a coarse vector to the fine lattice through them, and the coarse operator of a fine operator $X$ is its Galerkin projection $\mathcal{P}^\dagger X\mathcal{P}$. A few vectors per block suffice because low modes are locally coherent [20]. Finding them, the *setup*, is paid once per gauge field. The setup uses only the plain $D$ and $D_{\mathrm{PV}}$; Neff's $\alpha$ enters only the outer operator $\tilde{A}_L^{(\alpha)}$, the smoother that works on it, and the site-local factors of the correction (3.2), so one setup serves both forms and every $\alpha$.

### A. The factorized coarse-grid correction and the prolongation

#### 1. Why factorize

Solving on the even sites halves the problem and improves the conditioning, and red-black CG, the solver to beat, works there. But the Schur complement reaches two hops, so the existing methods make a choice (Table 1). HDCG solves on the even sites and coarsens the squared Schur complement, with an 81-point coarse stencil [38]. The methods with a small coarse stencil work on all lattice sites:

TABLE 1. Multigrid methods for domain-wall fermions: the system solved, the operator that is coarsened, and its coarse stencil, for blocks that contain the whole fifth dimension (except MG-PV, whose coarse lattice keeps the fifth dimension) and are at least two sites wide in each direction ($4^4$ for HDCG). $S$ is the Schur complement on the even sites.

| method | system solved | operator coarsened | coarse stencil |
|---|---|---|---|
| HDCG [38, 36] | even sites, $S^\dagger S$ | $S^\dagger S$ (four hops) | 81 points |
| HDCR [39] | all sites, $\gamma_5 RD$ | $\gamma_5 RD$, $c = 0$ (one hop) | 9 points |
| MG-PV [41, 40] | all sites, $D_{\mathrm{PV}}^\dagger D$ | $D$ and $D_{\mathrm{PV}}$ apart, $c = 0$ | 9 points per slice |
| Boyle and Oare [43] | all sites, $D_{\mathrm{PV}}^\dagger D$ | $D_{\mathrm{PV}}^\dagger D$ (two hops) | 33 points |
| this work | even sites, $\tilde{A}_R$ or $\tilde{A}_L$ | $D$ and $D_{\mathrm{PV}}$ apart (one hop each) | 9 points each |

TABLE 2. Site-local factors of the correction (3.2). They follow from $\tilde{A}^{-1}$ with $S^{-1} = \pi D^{-1}\iota$; with $\alpha$ the site-diagonal blocks are $C_e Q_e$ and $Q_e C_{\mathrm{PV},e}^\dagger$.

| form | $O_R$ | $O_M$ | $O_L$ |
|---|---|---|---|
| right, any $\alpha$ | $C_{\mathrm{PV},e}^\dagger$ | 1 | $C_e$ |
| left, $\alpha$ in $D$ only | 1 | $C_e Q_e C_{\mathrm{PV},e}^\dagger$ | $Q_e^{-1}$ |
| left, $\alpha$ in $D$ and $D_{\mathrm{PV}}$ | $Q_e^{-1}$ | $C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$ | $Q_e^{-1}$ |

HDCR [39] coarsens $\gamma_5 RD$, Boyle and Oare [43] coarsen $D_{\mathrm{PV}}^\dagger D$, and MG-PV [41, 40] coarsens $D$ and $D_{\mathrm{PV}}$ separately on a coarse lattice that keeps the fifth dimension. We keep the separate coarse operators of $D$ and $D_{\mathrm{PV}}$, use them for the even-odd operator, and make the coarse lattice four-dimensional.

The point counts follow from the hops. An operator that reaches $k$ hops on the fine lattice couples coarse sites whose coordinates differ by at most 1 in at most $k$ directions. For $k = 1$ and 2 this gives $1 + 8 = 9$ and $1 + 8 + 24 = 33$ points, and for $k \geq 4$ all $3^4 = 81$.

*The correction.* By $S^{-1} = \pi D^{-1}\iota$, the inverse of the right form is

$$\tilde{A}_R^{-1} = C_e\, S^{-1}\, (S_{\mathrm{PV}}^\dagger)^{-1}\, C_{\mathrm{PV},e}^\dagger = C_e\, \pi D^{-1}\iota\, \pi (D_{\mathrm{PV}}^\dagger)^{-1}\iota\, C_{\mathrm{PV},e}^\dagger, \tag{3.1}$$

a product of two inverses of one-hop operators and site-local factors. The Galerkin operator of $\tilde{A}_R$ itself would have a large coarse stencil, since $\tilde{A}_R$ reaches four hops. The Galerkin operators of $D$ and $D_{\mathrm{PV}}$ are nearest-neighbor, since $D$ is nearest-neighbor in four dimensions and every aggregate contains the whole fifth dimension. We therefore replace each inverse by its Galerkin approximation $\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger$. For both forms and any $\alpha$, the correction of a residual $r$ on the even sites is

$$e = O_L\, \pi \mathcal{P}\, \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^\dagger \iota \;\; O_M \;\; \pi \mathcal{P} \left(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}\right)^{-1} \mathcal{P}^\dagger \iota \;\; O_R\, r, \qquad \mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}, \tag{3.2}$$

with the site-local factors $O_R$, $O_M$, $O_L$ of Table 2. They come from writing each $\tilde{A}^{-1}$ as in Eq. (3.1). For example, for the left form with $\alpha$ in both operators, Eq. (2.26) gives

$$\tilde{A}_L^{(\alpha)\,-1} = Q_e^{-1}\, S^{-1}\, C_e Q_e^2 C_{\mathrm{PV},e}^\dagger\, (S_{\mathrm{PV}}^\dagger)^{-1}\, Q_e^{-1}, \tag{3.3}$$

so $O_R = Q_e^{-1}$, $O_M = C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$ and $O_L = Q_e^{-1}$. The two coarse inverses are those of the plain $D$ and $D_{\mathrm{PV}}$; within the correction, $\alpha$ is in these factors only. Figure 3 follows a residual through Eq. (3.2). It moves between the even sites, all lattice sites and the coarse lattice twice, and the result of the first coarse solve, taken on the even sites, is the right-hand side of the second.

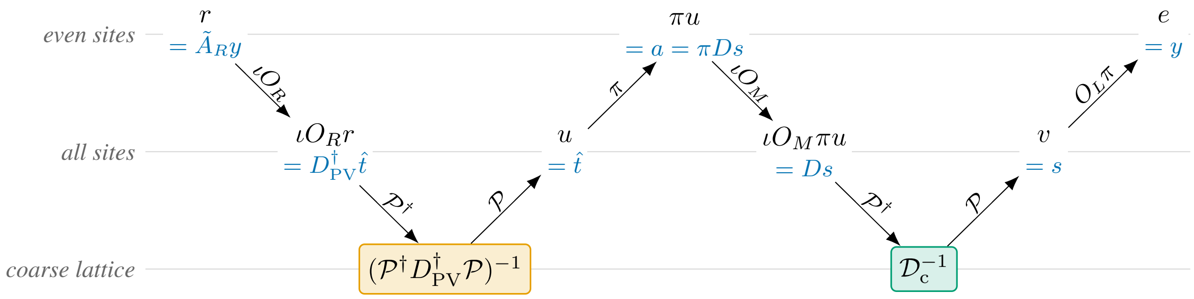

FIG. 3. One factorized coarse correction, from $r$ (left) to $e$ (right), which is Eq. (3.2) read from the right, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^\dagger D^\dagger_{\mathrm{PV}} \mathcal{P})^{-1} \mathcal{P}^\dagger \iota O_R r$ is the result of the first coarse solve, that of the adjoint Pauli–Villars operator, and its even part, with $O_M$, is the right-hand side of the second, that of the quark operator, whose result is $v = \mathcal{P} \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^\dagger \iota O_M \pi u$; $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Eq. (3.7); $O_R = C^\dagger_{\mathrm{PV},e}$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift $s$, and the correction returns $y$.

### 2. *Aggregation and prolongation*

We divide the fine lattice into rectangular four-dimensional blocks. Each block contains the whole fifth dimension and becomes one coarse site. We split each test vector $v$ into its $\Gamma_5$ eigencomponents $v_\pm = \frac{1}{2}(1 \pm \Gamma_5) v$ and orthonormalize the $+$ parts and the $-$ parts separately within each block. With $n_v$ test vectors a coarse site then carries $2n_v$ components: the first $n_v$ of positive and the last $n_v$ of negative $\Gamma_5$-chirality.

As in the chirality split of Wilson multigrid [28], each column of $\mathcal{P}$ is then an eigenvector of $\Gamma_5$. Since $\Gamma_5 = \gamma_5 R$, the condition $\Gamma_5 v_\pm = \pm v_\pm$ reads $v_\pm(L_s - 1 - s) = \pm \gamma_5 v_\pm(s)$. So each part is fixed by its values for $s < L_s/2$, and $\mathcal{P}$ is stored only there, at half the memory. On the even sites the transfer operators are $\mathcal{P}^\dagger \iota$ and $\pi \mathcal{P}$.

### 3. *The basis* $\{s, \hat{t}\}$

The prolongation is built from vectors $y_j$ on the even sites that approximate low modes of $\tilde{A}_R$ (Sec. III.C). Two facts shape how we use them.

- $\mathcal{P}$ and the coarse operators act on all lattice sites, so each $y$ must be extended, or *lifted*, to all lattice sites. We fill the odd sites so that the fine operator maps the lifted vector to the even-odd quantity, just as $\psi_o$ is rebuilt from $\psi_e$ after an even-odd solve.
- The correction (3.2) contains two Galerkin solves, one with $D^\dagger_{\mathrm{PV}}$ and one with $D$. Each is exact only on vectors in $\operatorname{range} \mathcal{P}$. So we lift each $y$ in two ways, one for each solve.

The two lifts are

$$s = \left(C_e^{-1} y,\ -C_o^{-1} D_{oe} C_e^{-1} y\right), \tag{3.4}$$

$$\hat{t} = \left(a,\ -(C^\dagger_{\mathrm{PV},o})^{-1} (D_{\mathrm{PV},eo})^\dagger a\right), \qquad a = \pi D s. \tag{3.5}$$

The lift $s$ is chosen so that $Ds$ vanishes on the odd sites. With the blocks of Eq. (2.22),

$$Ds = \begin{pmatrix} C_e C_e^{-1} y - D_{eo} C_o^{-1} D_{oe} C_e^{-1} y \\ D_{oe} C_e^{-1} y - C_o C_o^{-1} D_{oe} C_e^{-1} y \end{pmatrix} = \begin{pmatrix} S C_e^{-1} y \\ 0 \end{pmatrix} = \iota\, S C_e^{-1} y. \tag{3.6}$$

In the same way $\hat{t}$ is chosen so that $D^\dagger_{\mathrm{PV}} \hat{t}$ vanishes on the odd sites. The block form of $D^\dagger_{\mathrm{PV}}$ has $(D_{\mathrm{PV},eo})^\dagger$ as its odd-even block and $C^\dagger_{\mathrm{PV},o}$ as its odd diagonal block, and the same computation gives $D^\dagger_{\mathrm{PV}} \hat{t} = \iota\, S^\dagger_{\mathrm{PV}} a$.

The first coarse solve of the correction returns $\hat{t}$ and the second returns $s$ (Eq. (3.7) below). Since $\hat{t}$ is made from $Ds$, the two lifts follow the order of the solves. With the $y$ alone in the basis, each factor of the correction would be an approximation, and the errors of the two would multiply. The $2n$ vectors $\{s_j, \hat{t}_j\}$ are the test vectors, $n_v = 2n$.

The two lifts differ in the fifth dimension: their odd parts are built with $C_o^{-1}$ and with $(C_{\mathrm{PV},o}^{\dagger})^{-1}$, which by Eq. (2.17) carry each chirality toward opposite ends of the fifth dimension. So a space built from the $s_j$ alone does not contain the $\hat{t}_j$.

*Exactness for the right form.* Let $s, \hat{t} \in \operatorname{range} \mathcal{P}$, let $\mathcal{D}_{\mathrm{c}}$ and $\mathcal{P}^{\dagger} D_{\mathrm{PV}}^{\dagger} \mathcal{P}$ be invertible, and let both coarse systems in Eq. (3.2) be solved exactly, with the exact adjoint $\mathcal{P}^{\dagger} D_{\mathrm{PV}}^{\dagger} \mathcal{P} = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^{\dagger}$. Then the right-form correction returns the vector it is built from,

$$e(\tilde{A}_R y) = y. \tag{3.7}$$

For any invertible $\mathcal{P}^{\dagger} X \mathcal{P}$ the map $\mathcal{P}(\mathcal{P}^{\dagger} X \mathcal{P})^{-1} \mathcal{P}^{\dagger} X$ is the identity on $\operatorname{range} \mathcal{P}$, and we follow $r = \tilde{A}_R y$ through the correction with the factors of the right form, $O_R = C_{\mathrm{PV},e}^{\dagger}$, $O_M = 1$, $O_L = C_e$. First, $O_R r = C_{\mathrm{PV},e}^{\dagger} \tilde{A}_R y = S_{\mathrm{PV}}^{\dagger} S C_e^{-1} y = S_{\mathrm{PV}}^{\dagger} a$ and $\iota S_{\mathrm{PV}}^{\dagger} a = D_{\mathrm{PV}}^{\dagger} \hat{t}$, so the first coarse solve returns $\hat{t}$, whose even part is $\pi \hat{t} = a = \pi D s$. The second solve then has the right-hand side $\mathcal{P}^{\dagger} \iota \pi D s = \mathcal{P}^{\dagger} D s$, because $Ds$ vanishes on the odd sites, and returns $s$, whose even part is $\pi s = C_e^{-1} y$. Finally $O_L = C_e$ gives $e = y$.

*In practice.* Equation (3.7) holds only approximately: the stored $\mathcal{P}$ contains $s$ and $\hat{t}$ only up to the rounding of its reduced-precision storage, the coarse solves stop at a loose tolerance, and the Pauli–Villars solve is regularized. With both coarse factors solved exactly in double precision and $\mathcal{P}$ in half precision, the correction of $r = \tilde{A}_R y_j$ returns $y_j$ to a relative error of at most $2.5 \times 10^{-3}$ on $4^3 \times 8$ and $12^4$, against 3–37 for random vectors.

*The left form.* For the left form with $\alpha$ in both operators, the same argument needs the quark lift $s'$ of $Q_e y'$ and the Pauli–Villars lift $\hat{t}'$ of $u = C_{\mathrm{PV},e}^{\dagger\,-1} Q_e^{-2} C_e^{-1} S Q_e y'$ in $\operatorname{range} \mathcal{P}$ (Table 2 gives $O_M u = S Q_e y'$). For $\alpha = 1$ and $y' = C_e^{-1} y$ we have $s' = s$. But $u = C_{\mathrm{PV},e}^{\dagger\,-1} C_e^{-1} a$ differs from $a$ by a site-local matrix that mixes $s$ and spin, so $\hat{t}'$ is in general not in the span of the right-form vectors, and the correction is not exact for the left form. On $12^4$ it misses $y'$ by 1.6–24 in relative norm, not much less than random vectors (8–35). The left form converges nonetheless, with the spectrum of the right form (Sec. V.D.2), and it still needs $\hat{t}$ in the basis (Sec. V.C.2).

### 4. *The coarse operators*

We assemble $\mathcal{D}_{\mathrm{c}} = \mathcal{P}^{\dagger} D \mathcal{P}$ and $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^{\dagger} D_{\mathrm{PV}} \mathcal{P}$ from the hops of the fine operator between neighboring blocks and from its part within a block. Each coarse site has a $2n_v \times 2n_v$ diagonal block and one link block per direction, nine blocks in all. The adjoint factor $\mathcal{P}^{\dagger} D_{\mathrm{PV}}^{\dagger} \mathcal{P} = (\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}})^{\dagger}$ is the conjugate transpose of these blocks; the Pauli–Villars solve stores it as a second coarse operator. When all extents of the coarse lattice are even, the coarse hops connect coarse sites of opposite parity, and both coarse operators have an even-odd form of their own.

## B. The coarse solvers

We solve both coarse systems only approximately, in reduced precision, to a loose relative residual or a maximum number of iterations. Reaching the tolerance is not required. The two are solved in sequence: first the Pauli–Villars factor, whose solution enters the right-hand side of the quark factor. The default, used for all results unless stated, is BiCGStab on the even-odd form of the quark factor and CG on the regularized normal equation (3.9) of the Pauli–Villars factor, both to $|r|^2/|b|^2 < 10^{-2}$ (Algorithms 7 and 6, Appendix D).

### 1. *The quark factor*

Write $\mathcal{D}_\mathrm{c} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice. The system $\mathcal{D}_\mathrm{c} c = g$ is solved through its Schur complement,

$$\left(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}\right) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \qquad c_o = A_{oo}^{-1}(g_o - B_{oe} c_e), \tag{3.8}$$

with BiCGStab or truncated flexible GCR, preconditioned by the exact inverses of the diagonal blocks $A_{ee}^{-1}$. BiCGStab [22] is Algorithm 7 of Appendix D.

### 2. *The Pauli–Villars factor*

The system is $Xw = g$ with $X = (\mathcal{D}_\mathrm{c}^\mathrm{PV})^\dagger$. We solve the regularized normal equation

$$\left(X^\dagger X + \tau\right) w = X^\dagger g \tag{3.9}$$

by CG on all coarse sites, with $\tau = 10^{-3}$. The shift damps only the directions of $X$ whose singular values lie below $\sqrt{\tau}$. The Galerkin factor can have such directions although the fine $D_\mathrm{PV}$ does not (Sec. V.C.3).

### 3. *Levels below the coarse lattice*

The quark factor may be preconditioned by further levels. The coarse sites are grouped into super-blocks; on each super-block, local vectors are formed from the coarse test vectors $\mathcal{P}^\dagger s_j$ and $\mathcal{P}^\dagger \hat{t}_j$, split by chirality and orthonormalized (Gram–Schmidt applied twice); the level operators are obtained by probing the nearest-neighbor stencil; the last level is inverted densely. With $Q_\ell$ the prolongation into level $\ell - 1$ and $B_\ell$ the diagonal blocks of level $\ell$, we use a V-cycle with the coarse correction followed by one block-Jacobi sweep on the updated residual, without pre-smoothing,

$$z = Q_{\ell+1} M_{\ell+1} Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z); \tag{3.10}$$

on the coarse lattice itself the same multiplicative combination is used, and with the even-odd coarse form its even part preconditions the Schur system.

*Levels instead of deflation.* The levels below compress the low modes further by aggregation and invert the last level exactly, by LU in double precision (cuSOLVER on the device), so every mode that reaches that level is treated without computing any eigenvector. The alternative, deflating the coarse factors with a few of their eigenvectors, we have not tried with the basis $\{s, \hat{t}\}$. Boyle and Oare [43] use a large coarsest level that is inverted densely in the role of deflation.

## C. Setup

The setup produces the vectors $y_1, \ldots, y_n$ on the even sites, then the basis $\{s_j, \hat{t}_j\}$, $\mathcal{P}$ and the coarse operators. It always uses the right form; the same setup serves the left forms and every $\alpha$ (Sec. III.A). It has two stages. A Chebyshev filter finds approximate low modes without any multigrid. A bootstrap then improves them with the cycle built from them, so that they also cover what the first coarse space misses.

### 1. *Chebyshev-filtered subspace iteration*

The vectors approximate the right singular vectors of the smallest singular values of $\tilde{A}_R$, that is the low eigenvectors of the Hermitian positive operator $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. We find them by filtering. Let $T_d$ be the Chebyshev polynomial of degree $d$ and $\ell(x) = \big(2x - (\lambda_\mathrm{hi} + \lambda_\mathrm{lo})\big)/(\lambda_\mathrm{hi} - \lambda_\mathrm{lo})$. Then $|T_d(\ell(x))| \leq 1$ on $[\lambda_\mathrm{lo}, \lambda_\mathrm{hi}]$, while it grows quickly below $\lambda_\mathrm{lo}$. So $T_d(\ell(H_n))$ amplifies the low modes against the rest. We apply it by the three-term recurrence.

**Algorithm 1.** LowModes: Chebyshev-filtered subspace iteration on $H_n = \tilde{A}_R^\dagger \tilde{A}_R$.

```
$X \leftarrow n$ random vectors on the even sites
$\lambda_{\text{hi}} \leftarrow$ a bound slightly above $\lambda_{\max}(H_n)$ (power iteration);   $\lambda_{\text{lo}} \leftarrow$ a small fraction of $\lambda_{\text{hi}}$
for each round do
    $d \leftarrow$ a degree proportional to $\sqrt{\lambda_{\text{hi}}/\lambda_{\text{lo}}}$, within fixed bounds
    $X \leftarrow T_d(\ell(H_n))\, X$;   orthonormalize $X$ through its Gram matrix
    $G \leftarrow X^\dagger H_n X = V\Theta V^\dagger$;   $X \leftarrow XV$        ▷ Rayleigh–Ritz
    $\lambda_{\text{lo}} \leftarrow \theta_{\max}$        ▷ largest Ritz value
end for
return $X, \Theta$
```

### 2. Bootstrap

Each bootstrap round has three steps. It builds the basis and the two-grid cycle of Sec. III.D from the current $y$. It replaces every $y_j$ by a few FGMRES steps (Algorithm 5) on $\tilde{A}_R x = y_j$ from $x = 0$, preconditioned by that cycle. And it orthonormalizes the new $y$ through their Gram matrix ($y \leftarrow y V \Lambda^{-1/2}$ with $y^\dagger y = V \Lambda V^\dagger$). The cycle damps the components it already handles, so the new vectors are enriched in what it does not.

**Algorithm 2.** Setup of the two-grid method.

```
$y \leftarrow$ LowModes$(H_n)$
for each bootstrap round do
    build $\mathcal{P}$ from $\{s_j, \hat{t}_j\}$ of $y$ and the cycle $\mathcal{M}$ on it
    for $j = 1, \dots, n$ do
        $y_j \leftarrow$ a few FGMRES steps on $\tilde{A}_R x = y_j$, preconditioned by $\mathcal{M}$
    end for
    orthonormalize $y$
end for
compute $s_j, \hat{t}_j$ by Eqs. (3.4)–(3.5); split by $\Gamma_5$; orthonormalize per block $\rightarrow \mathcal{P}$
assemble $\mathcal{D}_{\text{c}}$, $\mathcal{D}_{\text{c}}^{\text{PV}}$, the inverses of their diagonal blocks, their even-odd forms
optionally, build the levels below from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$
```

## D. The Chebyshev smoother and the cycle

### 1. Smoother

The smoother is $\nu$ steps of Chebyshev iteration [48, 49] for $\tilde{A}x = b$ from $x = 0$, i.e. a fixed polynomial $x = p(\tilde{A})b$ of degree $\nu - 1$. It is defined by an ellipse with real foci $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$: $\lambda_{\text{hi}}$ slightly above an estimate of the spectral radius of $\tilde{A}$ from a power iteration, and $\lambda_{\text{lo}} = \lambda_{\text{hi}}/\kappa_s$ for a chosen ratio $\kappa_s$. With $\theta = \frac{1}{2}(\lambda_{\text{hi}} + \lambda_{\text{lo}})$, $\delta = \frac{1}{2}(\lambda_{\text{hi}} - \lambda_{\text{lo}})$, $\sigma = \theta/\delta$, $\varrho_0 = 1/\sigma$, $r_0 = b$ and $d_0 = r_0/\theta$, step $k$ is

$$x \leftarrow x + d_k, \quad r_{k+1} = r_k - \tilde{A} d_k, \quad \varrho_{k+1} = \frac{1}{2\sigma - \varrho_k}, \quad d_{k+1} = \varrho_{k+1}\varrho_k\, d_k + \frac{2\varrho_{k+1}}{\delta} r_{k+1}, \tag{3.11}$$

The last step only updates $x$, so $\nu$ steps cost $\nu - 1$ applications of $\tilde{A}$. The recurrence keeps four fine vectors, $x$, $r$, $d$ and $\tilde{A}d$, whatever $\nu$, and needs no inner products; the smoother is the same linear map at every application. A Krylov smoother, such as the fixed number of GCR steps of the Pauli–Villars multigrid of Ref. [40], keeps two more fine vectors per step, which in a fixed amount of memory lowers the largest volume that can be solved. We use only a real focal interval. The ratio $\kappa_s$ is not derived from the operator; it has to be chosen. The polynomial damps the part of the spectrum above $\lambda_{\mathrm{lo}}$ and leaves the rest to the coarse correction, and an interval that is too wide or too narrow can make the outer solve stagnate. The interval that works differs between the two setups of Sec. V (Secs. V.A and V.C.4).

### 2. *Cycle*

**Algorithm 3.** Two-grid cycle $y = \mathcal{M}b$ for $\tilde{A}$.

```
$y \leftarrow p(\tilde{A})\, b$                ▷ pre-smoothing
$r \leftarrow b - \tilde{A}y$
$y \leftarrow y + e(r)$                         ▷ correction (3.2)
$r \leftarrow b - \tilde{A}y$
$y \leftarrow y + p(\tilde{A})\, r$            ▷ post-smoothing
```

One cycle costs $2\nu$ applications of $\tilde{A}$ (two smoothings and two residuals) plus the two coarse solves. Because the coarse solves are Krylov iterations stopped by a residual test, $\mathcal{M}$ is not a fixed linear map.

## E. The outer solver

The outer iteration solves $\tilde{A}y = \rho$ with the cycle as a right preconditioner, and we rebuild $\psi$ as in Sec. II.C. The cycle $\mathcal{M}$ changes from one application to the next, so the outer method must be flexible: FGMRES or flexible GCR (Algorithms 5 and 4, Appendix D), which keep the preconditioned directions.

Flexibility alone gives no convergence guarantee. For a fixed $\mathcal{M}$, restarted or truncated methods converge for every restart length if the field of values of $\tilde{A}\mathcal{M}$ lies in the right half-plane [44]. Otherwise only the full method is covered, by the spectrum and the pseudospectra of $\tilde{A}\mathcal{M}$ [35, 50], and a truncated method can stagnate [51]. Secs. V.D.1–V.D.4 examine this for the present cycle. We monitor convergence on the even-odd system and evaluate the residual of the original equation, $\|D\psi - \eta\|/\|\eta\|$, with the plain $D$ after reconstruction. The reference solver is CG on the normal equation of the even-odd operator $C_e^{-1}S$ (CGNE). It is robust but squares the condition number. It runs in double precision, or in mixed precision: CG in single precision with reliable updates in double precision. Appendix D compares it with CG on $S^\dagger S$.

# IV. Solving with $D^\dagger$

HMC needs $(D^\dagger D)^{-1}$ at every force evaluation, so it solves with $D^\dagger$ as often as with $D$. The multigrid of $D$ also solves with $D^\dagger$, with no second setup, coarse operator or cycle. The construction needs only an approximate inverse of $D$, so it applies to any preconditioner built for the unsquared operator. The tool is the exact relation (2.13) between $D$ and $D^\dagger$.

## A. The idea

For Wilson fermions $D_W^\dagger = \gamma_5 D_W \gamma_5$, so $D_W^{-\dagger} = \gamma_5 D_W^{-1}\gamma_5$, and a solver for $D_W$ solves with $D_W^\dagger$ as well. For Möbius fermions this fails at non-zero $c$ (Sec. II.B.2). We use $\Gamma_5 D^{-1}\Gamma_5$ anyway, as a preconditioner for $D^\dagger$. The question is how good a preconditioner it is.

Multiply $D^\dagger$ by $\Gamma_5 D^{-1}\Gamma_5$ and use $D^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger$ from Eq. (2.13) and $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$D^\dagger\, \Gamma_5 D^{-1}\Gamma_5 = \Gamma_5 D_-^{-1} D\, D_- D^{-1}\, \Gamma_5 = \Gamma_5\big(1 + D_-^{-1}[D, D_-]D^{-1}\big)\Gamma_5. \tag{4.1}$$

The commutator follows from $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$: $[D, D_-] = -D_-[Y, D_-] = -c\,D_-[Y, W]$. The factor $D_-$ cancels, and

$$\boxed{D^\dagger\,\Gamma_5 D^{-1}\Gamma_5 = \Gamma_5\,(1-N)\,\Gamma_5, \qquad N = c\,[Y,W]\,D^{-1}.} \tag{4.2}$$

Equation (4.2) has three properties used below:

- $D_-^{-1}$ has dropped out. We never invert $D_-$, which we cannot do (Sec. IV.C).
- $[Y, W] = -\frac{1}{2}[Y, K]$ is a bounded operator: one hop in four dimensions and one in the fifth. So $N$ is large only where $D^{-1}$ is large, on the low modes of $D$.
- On a low mode $v$ the commutator acts through the $\gamma_\mu$ in $K$, which flip chirality. A low mode of $D$ carries on each wall the chirality bound there, $P_-$ at $s = 0$ and $P_+$ at $s = L_s - 1$; $[Y, W]v$ carries on each wall the opposite chirality, which no low mode has there. So $N$ takes the low modes out of their span, and $N^2$ is small. Then $(1 - N)^{-1} \approx 1 + N$, and a Krylov method needs only a few steps more than for the identity.

The third remark is a heuristic; Sec. V.B.2 tests it and finds that it fails on the smallest smeared lattice ($12^4$ in setup S).

### B. On the even sites

The solves of HMC are on the even sites, so we need the same idea for the Schur complement $S$. The blocks of $D^\dagger$ are $C_e^\dagger$, $C_o^\dagger$ on the diagonal and $D_{oe}^\dagger$, $D_{eo}^\dagger$ off it, so its Schur complement is $S^\dagger$, and $S^{-1} = \pi D^{-1}\iota$. Because $\Gamma_5$ acts within a site, it commutes with $\pi$ and $\iota$, and

$$S^{-\dagger} = \pi D^{-\dagger}\iota \approx \pi\,\Gamma_5 D^{-1}\Gamma_5\,\iota = \Gamma_5 S^{-1}\Gamma_5. \tag{4.3}$$

The solve of $D^\dagger w = r$ then goes as follows.

1. Form $r'_e = r_e - D_{oe}^\dagger (C_o^\dagger)^{-1} r_o$, so that $S^\dagger w_e = r'_e$.
2. Write $w_e = (C_e^\dagger)^{-1} z$. Then $\mathcal{K}^\dagger z = r'_e$ with $\mathcal{K} = C_e^{-1} S$, the operator HMC already uses.
3. Solve for $z$ by flexible GCR with the preconditioner $C_e^\dagger\,\Gamma_5 S^{-1}\Gamma_5$. The inverse $S^{-1}$ is one application of the cycle $\mathcal{M}$ of $\tilde{A}_R$: $S^{-1}t = C_e^{-1}\tilde{A}_R^{-1}(S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t \approx C_e^{-1}\mathcal{M}\,(S_{\rm PV} C_{{\rm PV},e}^{-1})^\dagger t$.
4. Recover $w_e = (C_e^\dagger)^{-1} z$ and $w_o = (C_o^\dagger)^{-1}(r_o - D_{eo}^\dagger w_e)$.

Each GCR step costs one cycle, as a step of the solve with $D$ does.

### C. Why $D_-$ cannot be inverted

Eq. (2.13) seems to offer more: the exact null vectors of $D^\dagger$, and an exact reduction of $D^\dagger w = r$ to a solve with $D$. Both need $D_-^{-1}$, and $D_-$ cannot be inverted in practice because of its spectrum. Since $D_- = cW - 1 = c\,(W - 1/c)$, $D_-$ is $c$ times the Wilson operator of mass $m_0 = -M_5 - 1/c$: $-3$ in setup S and $-3.8$ in setup B. The free eigenvalues of that operator,

$$m_0 + \sum_\mu (1 - \cos p_\mu) \pm i\Big(\sum_\mu \sin^2 p_\mu\Big)^{1/2}, \tag{4.4}$$

have real parts from $m_0$ to $m_0 + 8$, here from $-3$ to 5. Near real part zero they have imaginary parts of both signs, so the spectrum surrounds the origin. A Krylov method then has no low-degree polynomial $p$ with $p(0) = 1$ that is small on the spectrum (Sec. II.A), and it makes no progress. Numerically ($12^4$, stout, $M_5 = 1$), GCR(8) on $D_-$ and on $D_-^\dagger$ stops at $|r|^2/|b|^2 = 0.74$ (S) and 0.96 (B) after 20000 iterations, and BiCGStab diverges. The preconditioner of Eq. (4.2) avoids $D_-^{-1}$ altogether.

TABLE 3. The two setups and the multigrid of the mass scan. All runs: Möbius $(b, c) = (1.5, 0.5)$, $L_s = 8$, $m_{\mathrm{PV}} = 1$, $\alpha_{\mathrm{PV}} = 1$, boundary $(1, 1, 1, -1)$, $\alpha = 0.6$ unless stated. Configurations: quenched, plaquette action at $\beta = 6.0$, on $12^4$, $16^4$, $24^4$ and $24^3 \times 48$, four each; their plaquettes, $t_0$ and topological charges are in Appendix C.

| | setup S (stout) | setup B (bare) |
|---|---|---|
| $M_5$ | 1.0 | 1.8 |
| gauge smearing | stout, $\rho = 0.1$, 3 steps | none |
| Chebyshev smoother $\nu$, $\kappa_s$ | 8, 10 | 12, 30 |
| multigrid (both setups) | | |
| aggregates | $4^4$ ($24^3 \times 48$: levels below with super-blocks 2, 3) | |
| test vectors | 32 (16 filtered low modes of $\tilde{A}_R^\dagger \tilde{A}_R$, lifted to $\{s, \hat{t}\}$) | |
| setup | filter 2 rounds; bootstrap 2 rounds of 4 FGMRES steps | |
| coarse solves | single precision, $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-2}$; Pauli–Villars $\tau = 10^{-3}$ | |
| precision | $\mathcal{P}$ half, cycle single, outer solver double | |
| outer solver | GCR(2) on $\tilde{A}_L$ to $\lvert r\rvert^2/\lvert b\rvert^2 < 10^{-18}$ | |

## V. Numerical results

### A. Setups

*Numerical setup.* The quenched configurations use the plaquette action at $\beta = 6.0$, a lattice spacing of about 0.1 fm (Table 6). Results are means, medians or ranges over the four configurations of each volume as the captions state; the solvers without the multigrid and the pseudospectra use configuration 1. The two setups of the Möbius operator (Table 3) share $(b, c) = (1.5, 0.5)$, the choice of the JLQCD and the RBC/UKQCD Collaborations, and $L_s = 8$.

- *Setup B (bare links)* uses $M_5 = 1.8$, the value typical of simulations on unsmeared links [53].
- *Setup S (stout links)* smears the links of the fermion operator (stout, $\rho = 0.1$, three steps, Sec. II.B.5) and uses $M_5 = 1.0$, as in the JLQCD ensembles of Ref. [5].

Both keep the near-zero modes of $W = D_W(-M_5)$ small, by moving $M_5$ (B) or by smoothing the field (S), and so give the multigrid low modes of different character; they need different smoother intervals (Table 3). Timings are from one NVIDIA data-center GPU with 24 GB of device memory; a second GPU was used for iteration counts and spectral quantities only. The device reductions add in an order that depends on the build and the GPU, so the iterations of otherwise equal runs can differ by a few. The setup seed is 21 except in Sec. V.C.3.

### B. Performance

#### 1. Comparison with CGNE

On $24^3 \times 48$ the multigrid is 2.9–6.0 times faster than mixed-precision CGNE at $m = 0.001$, and its setup is recovered within one point-source propagator for $m \leq 0.005$. We compare first for one random right-hand side over seven masses, then for whole point-source propagators.

*The reference solver.* It is CG on $\mathcal{K}^\dagger\mathcal{K}$, $\mathcal{K} = C_e^{-1}S$, the even-odd operator of Bridge++, with Neff's $\alpha$ as for the multigrid, in mixed precision. Boyle and Yamaguchi [40] used CG on $S^\dagger S$, which here needs 21% (B) to 49% (S) more iterations (Appendix D).

*Mass dependence.* The multigrid removes most of the critical slowing down of CGNE (Fig. 4, Table 7). From $m = 0.1$ to $m = 0.001$ the iterations of CGNE grow about twentyfold, those of the multigrid by

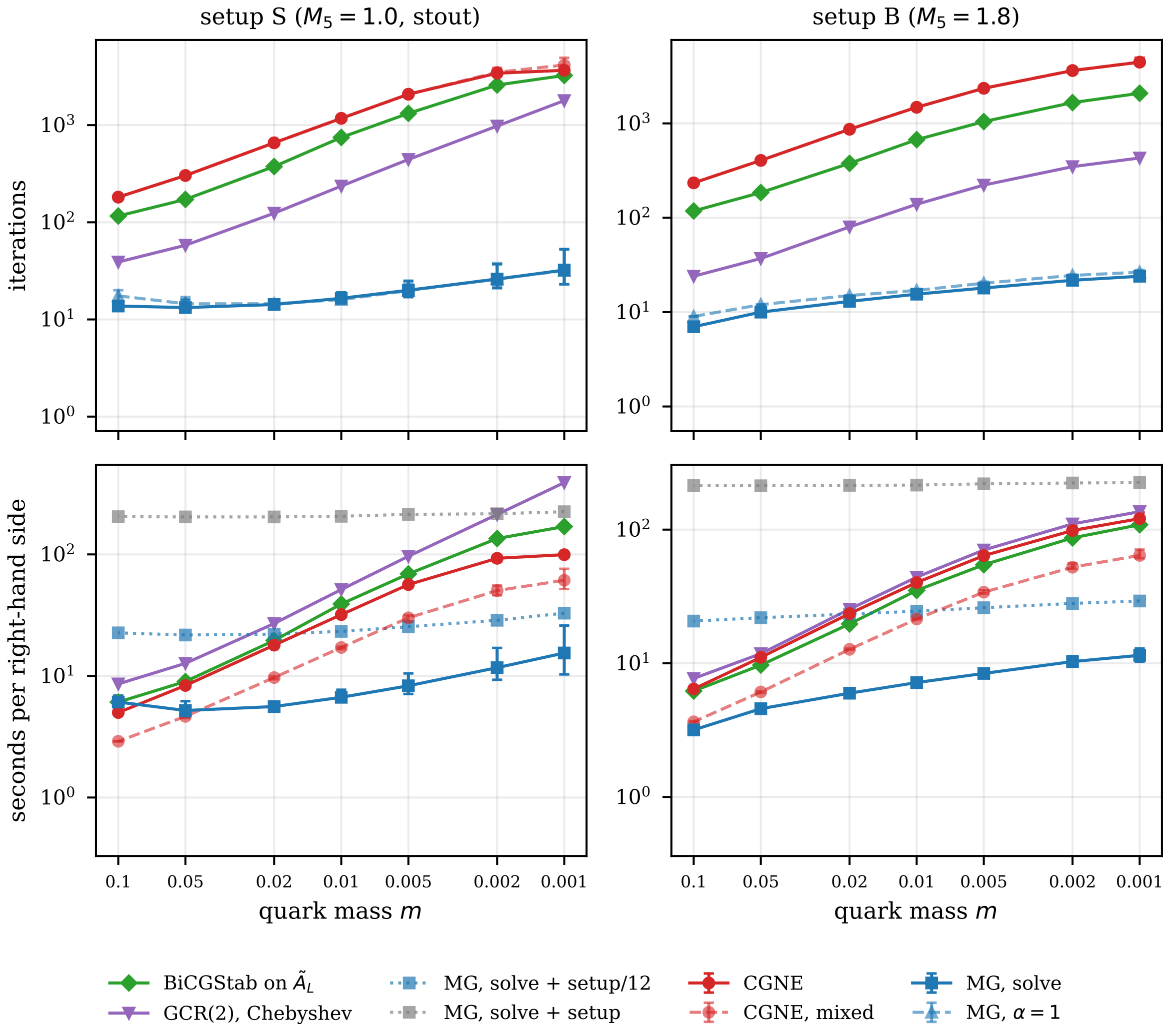

FIG. 4. The mass scan of Table 7, the mixed-precision CGNE, and the Krylov solvers on $\tilde{A}_L$ without the multigrid (Sec. V.C.1): iterations (top) and seconds per right-hand side (bottom) against the quark mass, $24^3 \times 48$. CGNE and multigrid: means over the four configurations, bars from the smallest to the largest value; the solvers without the multigrid: configuration 1. “Solve + setup/12”: the setup spread over the 12 columns of one point-source propagator.

1.6–4.1. Its setup of about 210 s is paid back after 3.5–5.2 right-hand sides at $m = 0.001$. For $m \geq 0.05$ in setup S, CGNE is faster or level. The multigrid needs three times the device memory of CGNE.

*A point-source propagator.* At $m = 0.001$ the multigrid takes 0.40–0.56 of the wall time of CGNE for a whole propagator, setup included (Table 4). At $m = 0.01$ the setup is not quite paid back within the 12 columns. The pseudoscalar correlators of the two solvers agree to $1 \times 10^{-6}$ or better.

*Volume.* The multigrid solve is 1.0–2.1 times as fast as CGNE on $16^4$ and 2.9–6.0 times from $24^4$ on; on $12^4$ it is slower (Fig. 5, Table 9). Its iterations stay at 12–53 from $12^4$ to $24^3 \times 48$, apart from one slow setup on $12^4$, while those of CGNE grow 8–16 times. Its setup grows with the volume.

*Neff’s $\alpha$.* At $m = 0.001$, $\alpha = 0.6$ against $\alpha = 1$ saves CG 25–35% of its iterations and the multigrid at most two (Table 7), as Sec. II.C expects.

### 2. *Solves with $D^\dagger$*

*Single solves.* Table 10 compares three ways to solve with $D^\dagger$ from one setup of $D$. The right-hand sides are a random vector and the residual $r = b - D^\dagger D x_0$ after one sweep of the normal equations, solved to

TABLE 4. Point-source propagators (12 columns) on $24^3 \times 48$: the multigrid (setup seconds, iterations and seconds per column) against the mixed-precision CGNE (iterations and seconds per column), the wall time of the whole propagator, and the largest relative difference of the pseudoscalar correlator at $t = 0 \dots 3$. At $m = 0.001$: ranges over the four configurations and the 12 columns; at $m = 0.01$: configuration 1.

| | | MG | | | CGNE, mixed | | wall (s) | | |
|---|---|---|---|---|---|---|---|---|---|
| | $m$ | setup | it | s | it | s | MG | CGNE | $\lvert\Delta C\rvert/C$ |
| S | 0.001 | 201–216 | 26–59 | 11.7–29.4 | 3634–6400 | 52.1–91.6 | 381–562 | 678–1048 | $\leq 1 \times 10^{-6}$ |
| B | 0.001 | 213–215 | 23–30 | 10.7–14.8 | 4354–5574 | 62.4–79.8 | 372–409 | 803–973 | $\leq 7 \times 10^{-7}$ |
| S | 0.01 | 203 | 16–17 | 6.4–7.0 | 1337–1388 | 19.8–20.5 | 305 | 264 | $2 \times 10^{-7}$ |
| B | 0.01 | 215 | 17 | 8.1 | 1641–1745 | 24.2–25.7 | 331 | 324 | $4 \times 10^{-8}$ |

$10^{-10}$. The second kind is the one that matters in HMC: its error lies mostly on the lowest modes.

- *The cycle of $\tilde{A}_R^\dagger$ on the same $\mathcal{P}$* (the "dual cycle"): the factorized correction (3.2) with $D$ and $D_{\mathrm{PV}}$ exchanged.
- *Flexible GCR on $D^\dagger$ with* $\Gamma_5 \mathcal{M} \Gamma_5$, Eq. (4.2), one cycle a step.

On $16^4$ the preconditioner $\Gamma_5 \mathcal{M} \Gamma_5$ needs two to three iterations more than the solve with $D$, in both setups and on both kinds of right-hand side, as the heuristic predicts. In setup S more inner cycles per step do not reduce the count, so the extra steps come from $N$ and not from the inexact $\mathcal{M}$. The dual cycle stalls on the residual right-hand side in setup S on $12^4$ and on three of the four $16^4$ configurations.

On $12^4$ in setup S, $m = 0.001$, the residual right-hand side stalls GCR(2) on all four configurations, and GCR(8) needs 13–222 iterations. On $16^4$, where configurations 2–4 have $Q = +1$, $+1$ and $+3$, GCR(2) needs 9–10 on all four. The stall follows the small smeared lattice of Sec. V.D.4, which we infer from the coincidence only: no $12^4$ run without smearing was made for $D^\dagger$. The topological charge does not predict it. On the same right-hand side the dual cycle does not converge on $12^4$ in setup S.

*What did not work.* None of the other ways we tried to solve with $D^\dagger$ from one setup is as good as $\Gamma_5 \mathcal{M} \Gamma_5$. The exact map of Eq. (2.13) needs $D_-^{-1}$, which does not converge (Sec. IV.C). For the dual cycle, $\mathcal{P}$ from $y \cup \Gamma_5 y$ converges but quadruples the coarse operators, other prolongations need up to 300 iterations, and GCR(4) in place of BiCGStab for its Pauli–Villars factor converges at three times the cost per cycle.

*What these solves are for.* In HMC each force evaluation needs one solve with $D$ and one with $D^\dagger$, and with $\Gamma_5 \mathcal{M} \Gamma_5$ one setup serves both. The cost in HMC is left to future work (Sec. VII).

### C. What each part contributes

The speed-up of Sec. V.B.1 rests on four choices beyond the operator $A'$: the coarse correction, the lift $\hat{t}$ in the basis, the regularized Pauli–Villars solve, and the setup parameters. Each part below isolates one of them.

#### 1. Krylov on $\tilde{A}_L$ without the multigrid

The speed-up comes from the coarse correction: without it a Krylov method on $\tilde{A}_L$ does not beat CGNE (Fig. 4). BiCGStab on $\tilde{A}_L$ alone needs about as many iterations as double-precision CGNE, but each of its iterations applies $\tilde{A}_L$ twice, and at $m = 0.001$ in setup S it takes 1.7 times the time of double-precision CGNE. The Chebyshev smoother as a preconditioner cuts its iterations but raises the time.

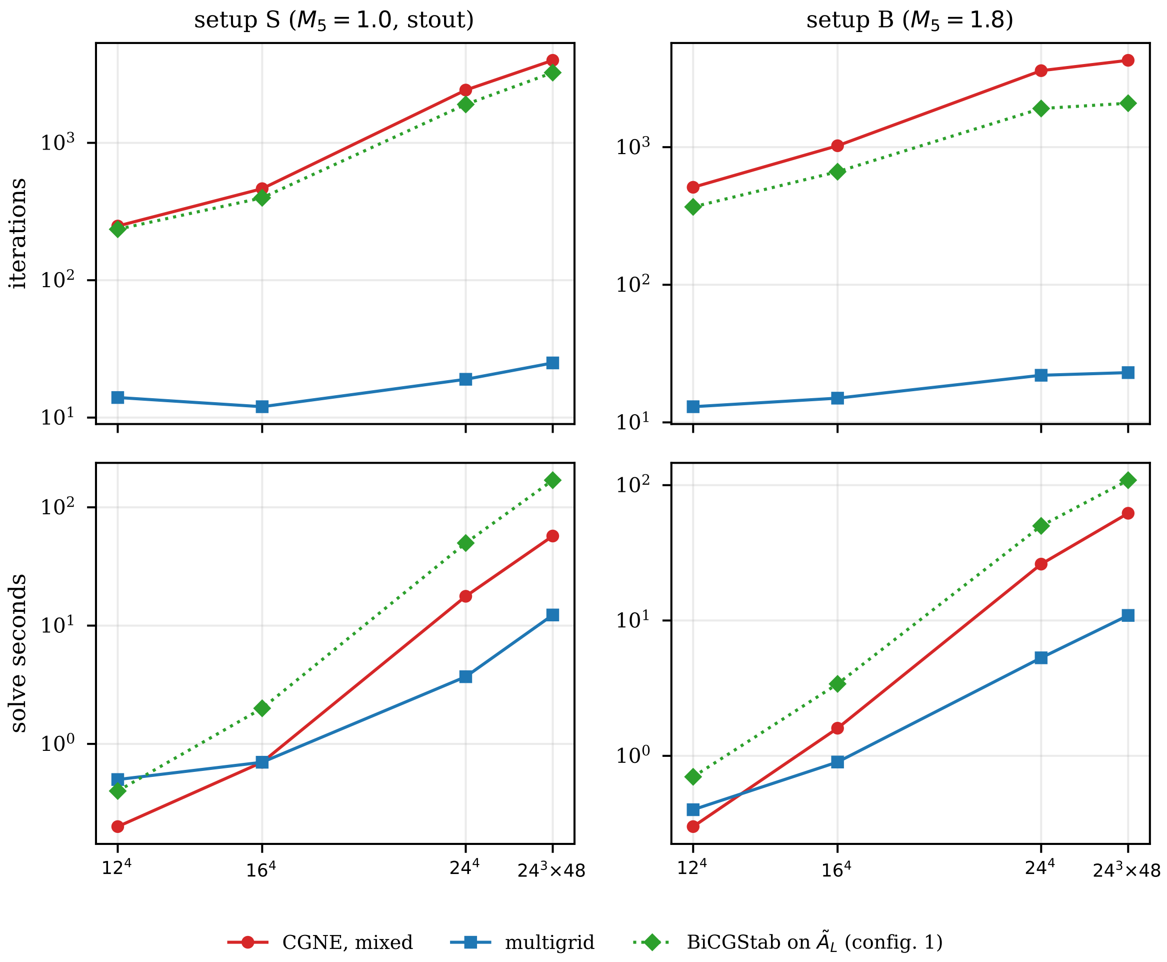

FIG. 5. Iterations (top) and solve seconds (bottom) at $m = 0.001$ against the volume, one random right-hand side. Mixed-precision CGNE and multigrid: median over the four configurations; BiCGStab on $\tilde{A}_L$: configuration 1. The median hides the slow setup on $12^4$ in setup S (Sec. V.D.4); the ranges are in Table 9.

### 2. *Choice of the prolongation basis*

The exactness (3.7) needs both lifts $s$ and $\hat{t}$ of each $y$ in range $\mathcal{P}$, but the left form used in practice is not exact (Sec. III.A.3). To see what $\hat{t}$ does there, we replace it by $Ds$, which vanishes on the odd sites, or drop it (Table 11). With $\{s\}$ alone $\mathcal{P}$ has half the columns; to compare at the same size we also take 32 vectors $y$, which doubles the vectors of the filter and of the bootstrap.

With the same vectors $y$, the two lifts give the fewest iterations on every volume. In setup S, replacing $\hat{t}$ by $Ds$ or dropping it stagnates on three of the four $12^4$ and $16^4$ configurations and costs five times the iterations on $24^3 \times 48$; in setup B it costs 1.2 to 2.2 times the iterations. Twice as many vectors $y$ with $\{s\}$ alone take the place of $\hat{t}$ on $24^3 \times 48$ but not on $12^4$, at 1.6–1.8 times the setup time and twice the host memory; a point-source propagator is still faster with the two lifts in both setups.

### 3. *The regularized Pauli–Villars solve*

With the regularized solve no slow setups remain on $12^4$: all 40 setups of setup S need 10–17 iterations, against 70 to more than 2000 for the slow ones without it (Table 12). The cause is the Galerkin factor. With stout smearing its smallest singular value lies one to two orders of magnitude below that of the fine $D_{\mathrm{PV}}$; without smearing the two agree (Table 13).

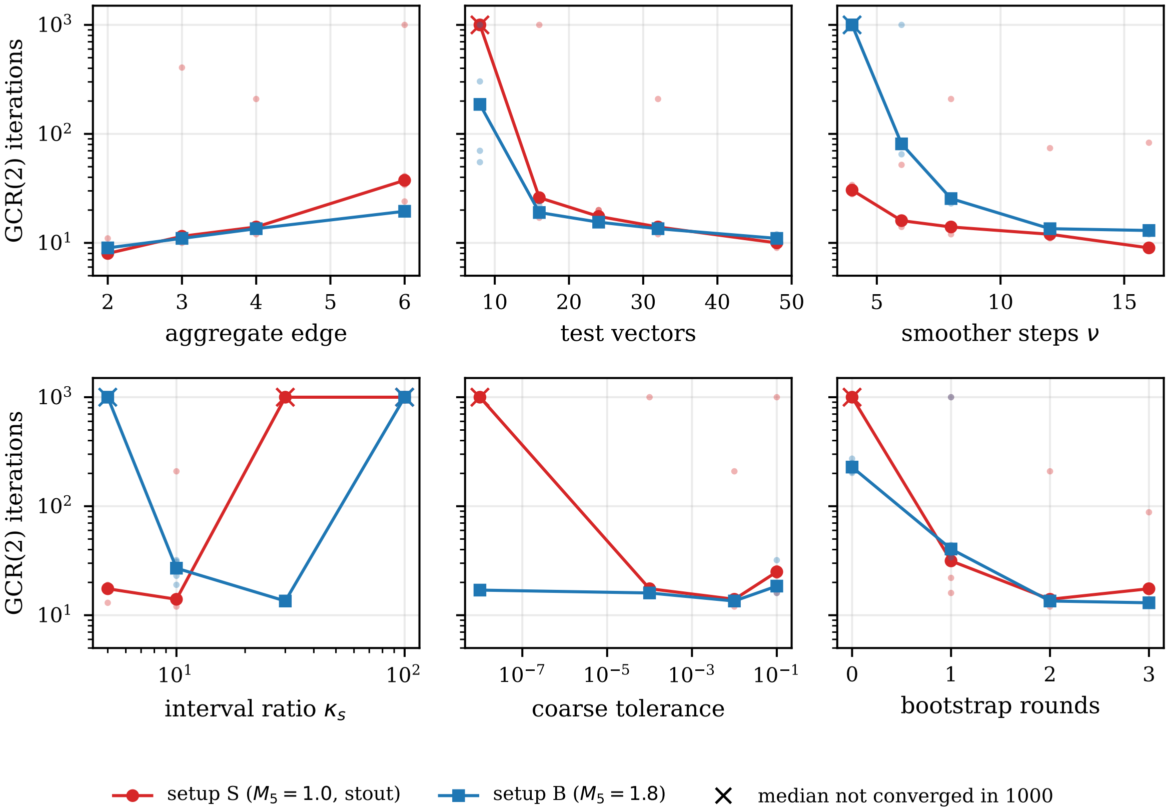

FIG. 6. GCR(2) iterations on $\tilde{A}_L$ when one parameter of Table 3 is changed, $12^4$, $m = 0.001$, one random right-hand side. Lines: median over the four configurations; dots: the four configurations; a cross marks a median that does not converge in 1000 iterations. The default is the point where the lines of the two setups pass through 14 iterations. Values and setup and solve times in Table 14.

#### 4. *Sensitivity to the parameters*

The method is robust within a window of each parameter and fails outside it (Fig. 6). The smoother interval must suit the setup, the bootstrap is necessary, and more accurate coarse solves do not help.

Fewer iterations are not always less time (Table 14). $2^4$ aggregates need 8–9 iterations, but the larger coarse lattice makes the solve and the setup two to three times as long as with $4^4$; 48 test vectors likewise save iterations and lose in setup and time per iteration. On $24^3 \times 48$, 16 test vectors do not suffice: the solve needs hundreds of iterations or does not converge in 2000.

Whyte, Stathopoulos and Romero [54] obtain, for Wilson fermions, the effect of more test vectors at the coarse size of fewer by truncating them with a singular value decomposition on each aggregate. Here the truncated basis would no longer contain $s$ and $\hat{t}$ exactly; whether the gain outweighs that loss is open.

### D. Why the method converges

#### 1. *Field of values: the coarse correction amplifies*

The method converges as in Sec. V.B.1 although the usual theory does not apply. The four parts of this subsection ask which property of the preconditioned operator sets the convergence. Two notions from non-Hermitian linear algebra are used. The *field of values* $W(X) = \{z^\dagger X z / z^\dagger z\}$ contains the spectrum, and its leftmost point is the smallest eigenvalue of the Hermitian part $(X + X^\dagger)/2$; GCR is guaranteed to converge at every restart length when $W$ lies in the right half-plane [44]. The $\varepsilon$*-pseudospectrum* is the set of $z$ where $\sigma_{\min}(z - X) < \varepsilon$, i.e. where $\|(z - X)^{-1}\| > 1/\varepsilon$; for a normal $X$ it is the $\varepsilon$-neighborhood of the spectrum, for a non-normal one it can be much larger, and it bounds the convergence of full GMRES [50].

The condition number of an eigenvalue, $\kappa = 1/|y^\dagger x|$ for its right and left eigenvectors $x$ and $y$ normalized to $\|x\| = \|y\| = 1$, is 1 for a normal matrix; a large $\kappa$ means that the eigenvalue moves by $\kappa\varepsilon$ under a perturbation of size $\varepsilon$, which widens the pseudospectrum around it (Secs. V.D.2, V.D.3).

The coarse correction amplifies some vectors by $10^3$–$10^5$, and the field of values of the preconditioned operator reaches into the left half-plane (this subsection), so the bound of Eisenstat, Elman and Schultz [44] for GCR is lost. Secs. V.D.2, V.D.3 and V.D.4 give what remains: the eigenvalues, the pseudospectra and the truncated GCR used in practice.

The Hermitian part of $\tilde{A}_R\mathcal{M}$ was computed with $\mathcal{M}$ made a fixed linear map: both coarse factors solved to $|r|^2/|b|^2 < 10^{-20}$ in double precision with the exact adjoints, and $\mathcal{M}^\dagger$ built with the same $\mathcal{P}$ and the same smoother interval, so that $\langle z|\tilde{A}_R\mathcal{M}w\rangle$ and $\langle \mathcal{M}^\dagger \tilde{A}_R^\dagger z|w\rangle$ agree to $10^{-10}$. The cycle was also taken apart: the smoother alone, $\mathcal{S} = p(\tilde{A}_R)$, the coarse correction alone, $E$ of Eq. (3.2), and the cycle $\mathcal{M}$ (Table 15).

The coarse correction is what pushes the field of values into the left half-plane. On $4^3 \times 8$ the field of values of $\tilde{A}_R\mathcal{M}$ lies in the right half-plane, whereas on $12^4$ and $16^4$ it does not. Table 15 shows where the change comes from.

- The field of values of $\tilde{A}_R$ itself hardly depends on the volume. From $12^4$ to $24^4$ its upper edge moves by less than 1%, and its lower edge goes from $-4.6 \times 10^{-3}$ to $-1.3 \times 10^{-2}$ (S) and stays at $-5 \times 10^{-2}$ (B).
- The smoother is harmless: $\|I - \tilde{A}_R\mathcal{S}\| \approx 1$.
- The coarse correction amplifies: $\|I - \tilde{A}_R E\|$ is $10^3$–$10^5$ (up to $3 \times 10^5$ on the four $12^4$ configurations), against 18.5 on $4^3 \times 8$. $\tilde{A}_R E$ is the identity on the vectors the correction is built from (Eq. (3.7)), but it amplifies other directions by three to four orders of magnitude, like a strongly oblique projection.

Eick and Mawhinney [52] describe the same mechanism for a renormalization-group coarse lattice. The coarse inverse is large on the low modes of the coarse operator. The interpolated correction is not exactly a low mode of the fine operator, and the part that misses comes back amplified. In their case the fine operator is $D^\dagger D$, whose spectrum reaches 520 against 0.05 for its lowest modes; it supplies the four orders of magnitude, and they need a Chebyshev filter of degree 150–300. In our case $\tilde{A}_R$ is close to the identity on the high modes, because the Pauli–Villars operator divides them out, and its field of values ends at 2 (S) and 4.5 (B). The large factor is the coarse inverse itself. The post-smoothing, a Chebyshev polynomial of degree 7–11 on the upper part of the spectrum, acts as the filter, and the whole cycle keeps $\|I - \tilde{A}_R\mathcal{M}\|$ at 55 (S) and 5.6 (B) on configuration 1, and at 15–722 over the four configurations in S.

So the GCR bound of Ref. [44] does not apply. The field of values is also not what decides the iteration count. If we replace $E$ by the Galerkin correction of $\tilde{A}_R$ itself, $\mathcal{P}(\mathcal{P}^\dagger\tilde{A}_R\mathcal{P})^{-1}\mathcal{P}^\dagger$ (built densely), the field of values of setup B moves into the right half-plane, $[0.121, 1.99]$, and $\|I - \tilde{A}_R\mathcal{M}\|$ drops to 1.84 (setup S: $[-3.62, 13.1]$, 16.8). Yet GCR(8) to $10^{-16}$ then needs more iterations: 16 against 13 (B) and 12 against 10 (S).

### 2. *Spectrum: far from the origin*

The spectrum, unlike the field of values, stays far from the origin. Every eigenvalue lies at $\mathrm{Re}\,\lambda \geq 0.69$, with moderate condition numbers (Table 16, Fig. 7). On configuration 1 setup S has one ($12^4$) or two ($16^4$) isolated outliers of large modulus, and all other eigenvalues lie within 0.43–0.47 of 1; on configurations 2–4 of $16^4$ the eigenvalue farthest from 1 lies at $|\lambda| = 2.7$–$4.6$. The left and right forms have the same spectrum to the precision shown. At $m = 0.005$ on $16^4$ the lowest real parts are 0.802 (S) and 0.790 (B).

### 3. *Pseudospectra and a bound for GMRES*

For a non-normal operator the eigenvalues alone do not bound the convergence of GMRES; the $\varepsilon$-pseudospectrum, the set where $\sigma_{\min}(z - X) < \varepsilon$, does [35, 50]. If a contour $\Gamma$ of length $L$ encloses the

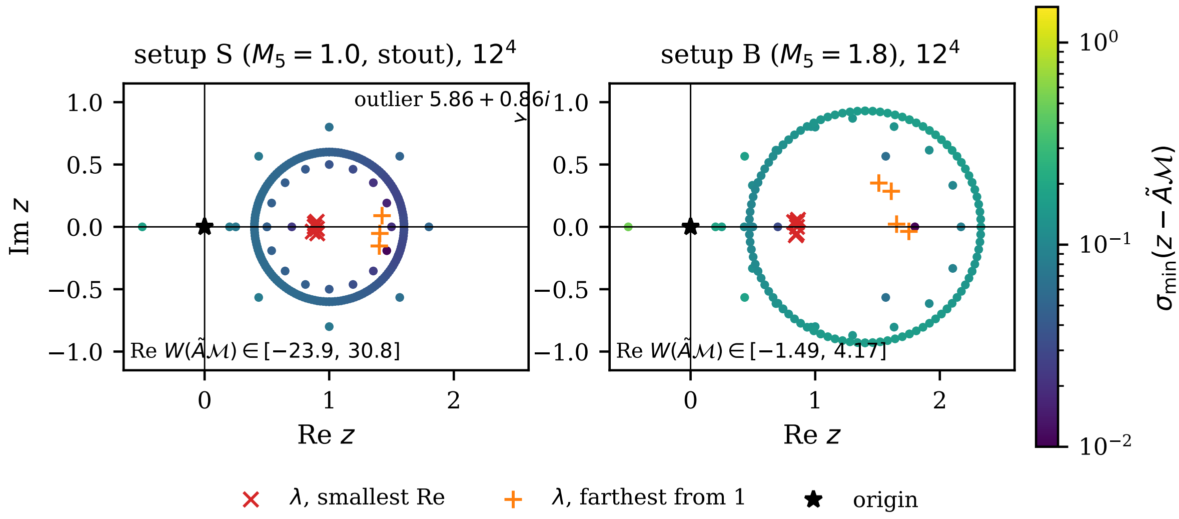

FIG. 7. $12^4$, $m = 0.001$, exact coarse solves: the six eigenvalues of $\tilde{A}_R\mathcal{M}$ of smallest real part, the four farthest from 1, and $\sigma_{\min}(z - \tilde{A}_R\mathcal{M})$ (color) at the points where it was computed: the real axis, the contours of Table 17 and the dense circles $|z-1| = 0.6$ (S) and $|z - 1.4| = 0.93$ (B). The field of values reaches far into the left half-plane (its real range in the corner), the spectrum does not.

spectrum of $X$ but not the origin and $\sigma_{\min}(z - X) \geq \varepsilon$ on it, then for any polynomial $p_k$ with $p_k(0) = 1$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \|p_k(X)\| \leq \frac{L}{2\pi\varepsilon} \max_{z\in\Gamma} |p_k(z)|. \tag{5.1}$$

An isolated outlier $\lambda_o$ outside $\Gamma$ is removed by the factor $(1 - z/\lambda_o)$ of $p_k$: then $p_k(X) = p_k(X)(I - P_o)$ with $P_o$ the spectral projector, $\|(z-X)^{-1}(I - P_o)\| \leq \kappa_o/\varepsilon$, and with a circle $|z - c| = r$ and $p_k(z) = (1 - z/\lambda_o)(1 - z/c)^{k-1}$,

$$\frac{\|r_k\|}{\|r_0\|} \leq \kappa_o \, \frac{r}{\varepsilon} \, \frac{|\lambda_o - c| + r}{|\lambda_o|} \left(\frac{r}{c}\right)^{k-1}. \tag{5.2}$$

On $16^4$ setup S has two outliers; the factor $\kappa_o$ is then replaced by $\|I - P_1 - P_2\| \leq 1 + \kappa_1 + \kappa_2$ and there are two factors of the form $(|\lambda_o - c| + r)/|\lambda_o|$. On the real axis between the origin and the spectrum, $\sigma_{\min}(z - \tilde{A}_R\mathcal{M}) \geq 0.024$ (S) and $\geq 0.031$ (B) on $12^4$ ($z = -0.5 \ldots 0.7$), and 0.107 (S), 0.271 (B) at the origin.

The bound guarantees convergence at a fixed rate, $r/c$ per iteration, with no stagnation. It is pessimistic by a factor of 3–5 in iterations: GCR needs 10–17 with exact coarse solves (Sec. V.D.4). Its rate, however, is the one we measure on $24^3 \times 48$ on configurations 1 and 3. There the median factor per GCR step of $|r|^2$ is 0.23–0.34 (Fig. 8), that is 0.48–0.58 in $|r|$, against $r/c = 0.5$–$0.6$ for setup S. Configuration 2 in setup S converges more slowly, at a steady 0.56 per step in $|r|^2$. Setup B does not get worse from $12^4$ to $16^4$ ($\min \sigma$ grows from 0.098 to 0.13). In setup S the outliers of configuration 1 grow, but their condition numbers fall and $\varepsilon$ stays the same.

*What is not yet rigorous.* On the two dense circles of $12^4$ the spacing of the points is below $\varepsilon$, and since $\sigma_{\min}(z - X)$ is 1-Lipschitz in $z$ the bound holds on the whole circle, $\sigma_{\min} \geq 0.0164$ (S) and $\geq 0.0676$ (B), with 54 and 58 guaranteed iterations. The circles of $16^4$ have only 16 points. The Lanczos estimates approach $\sigma^2_{\min}$ from above, the disc rests on the eigenvalues farthest from $c$ found by Arnoldi, and the bound is for full GMRES with a fixed $\mathcal{M}$ (Secs. V.D.4, VII).

### *4. Truncated GCR: inexact coarse solves*

Truncated GCR does not need accurate coarse solves, and with them it can fail. With the usual coarse solves to $10^{-2}$, GCR(2) needs one to five iterations more than the full method on $12^4$ (Table 18). With coarse solves to $10^{-8}$, where $\mathcal{M}$ is close to the fixed map of Secs. V.D.1–V.D.3, GCR(2) stagnates in setup S on all four configurations ($|r|^2 \approx 3 \times 10^{-8}$ after 1000 iterations on configuration 1), while GCR(8) converges in 17–26 on three of them and GCR($\geq 16$) in 15–17 on all four. The pseudospectra guarantee the full method; a truncated method has no guarantee when the field of values contains the origin [44, 51]. With the exact cycle (double precision, $10^{-20}$) GCR(2) stagnates only on configurations 1 and 4 of $12^4$; on $16^4$ and $24^4$ it converges (Table 20).

Inexact coarse solves do not keep GCR(2) converging on $12^4$ by leaving out the directions in which $E$ amplifies. Take the vector $x$ at the left end of the field of values of $\tilde{A}_R\mathcal{M}$ (exact coarse solves, $12^4$). With coarse solves to $10^{-2}$ the cycle gives $\mathrm{Re}\; x^\dagger \tilde{A}_R \mathcal{M} x = -25.2$ (S) and $-1.43$ (B), against $-24.3$ and $-1.49$ with exact ones, and $\|(I - \tilde{A}_R E)x\| = 1.16 \times 10^3$ and $1.03 \times 10^2$, against $1.13 \times 10^3$ and $1.12 \times 10^2$. With tolerances $10^{-8}$ and $10^{-4}$ they agree with the exact solves to within $1\%$. What the tolerance changes is that the cycle is no longer a fixed linear map: with Krylov coarse solves stopped early, it depends on its input.

*How GCR(2) stagnates.* GCR(2) with a fixed preconditioner can reach a fixed point. On $12^4$ in setup S, $|r|^2$ falls for about 15 steps and then does not change in any digit for the next 480. Meanwhile $|\langle r|\tilde{A}_L\mathcal{M}r\rangle|/(\|r\|\,\|\tilde{A}_L\mathcal{M}r\|)$ falls from about 0.8 to $10^{-10}$–$10^{-9}$. The new direction is then orthogonal to the residual, and since $\mathcal{M}$ is fixed the next step repeats it. A field of values that contains the origin admits such a residual. Any variation of the cycle leaves the fixed point: on configuration 1 of $12^4$ the exact cycle and two fixed perturbations of it stagnate in all 15 runs, and random perturbations, down to $2\%$ of the cycle, converge in all 75 (Table 20). On the four quenched $16^4$ configurations, on the dynamical one and on $24^4$ the exact cycle does not reach a fixed point in the runs made. A fixed perturbation can still produce one there ($\delta\Gamma_5\mathcal{M}r$ on $24^4$), so whether a fixed cycle stagnates depends on the cycle and on the lattice. Of the four $12^4$ configurations, 1 and 4 reach it and 2 and 3 do not, so the fixed point depends on the configuration; which property of it decides is open. It is neither the topological charge (configurations 3 and 4 both have $Q = +2$, and $24^4$ has $Q = +4$ and converges) nor the odd extent of the $3^4$ coarse lattice ($3^4$ aggregates give a $4^4$ coarse lattice and stagnate as well).

*Dependence on the setup seed.* On $12^4$ with stout smearing about one setup in eight makes GCR(2) slow. The seed of the random starting vectors that the setup filters (21 in all other runs) changes the coarse space but not the configuration, the operator or the right-hand side, and the solution agrees to the stopping tolerance. Table 19 repeats the solve at $m = 0.001$ with many seeds, on the four configurations of Table 6 for $12^4$. In setup S, 19 of 160 setups need more than 30 iterations and 13 of them more than 200 or do not converge in 2000. This is 2 to 7 per configuration, which does not distinguish the configurations at these counts. No setup is slow without smearing at $M_5 = 1$, in setup B, on $16^4$ (quenched and dynamical), or on $24^3 \times 48$ configurations 3 and 4, where five seeds need 18–28 iterations. Configuration 2 of $24^3 \times 48$ needs 36–53 iterations with all three seeds, against 18–28 on the other three, so that excess belongs to the configuration and not to the seed.

A slow setup gives a cycle of the usual quality, and GCR(2) fails to use it. With the same cycles GCR(16) needs 11–14 iterations in all 40 setups tested. In the slow runs $|r|^2$ falls normally for a few steps and then stays nearly constant, for up to 300 steps at $|r|^2/|b|^2 \approx 7 \times 10^{-10}$ on configuration 1, before it falls again; this is the behavior near a fixed point of Table 20, reached here with inexact coarse solves. The slowness belongs to the setup: five of the six slow setups of the first ten seeds are slow for the point source as well (44 to more than 500 iterations). At $L_s = 12$ and 16 the same occurs, in 7 of 20 setups on configurations 2 and 4.

Smearing moves the field of values of the cycle across the origin on $12^4$. With exact coarse solves and stout smearing, $\mathrm{Re}\, W(\tilde{A}_R\mathcal{M})$ reaches $-6.3$ to $-324$ on the four configurations; with $M_5 = 1$ and no smearing it lies in $[0.23, 2.62]$ on all four (Table 15). The field of values of $\tilde{A}_R$ itself is the same in both cases. The difference is the coarse correction: $\|I - \tilde{A}_R E\|$ is $4.7 \times 10^3$–$2.8 \times 10^5$ with smearing and 64–84 without. With the origin outside the field of values, truncated GCR($k$) converges for every $k$ [44],

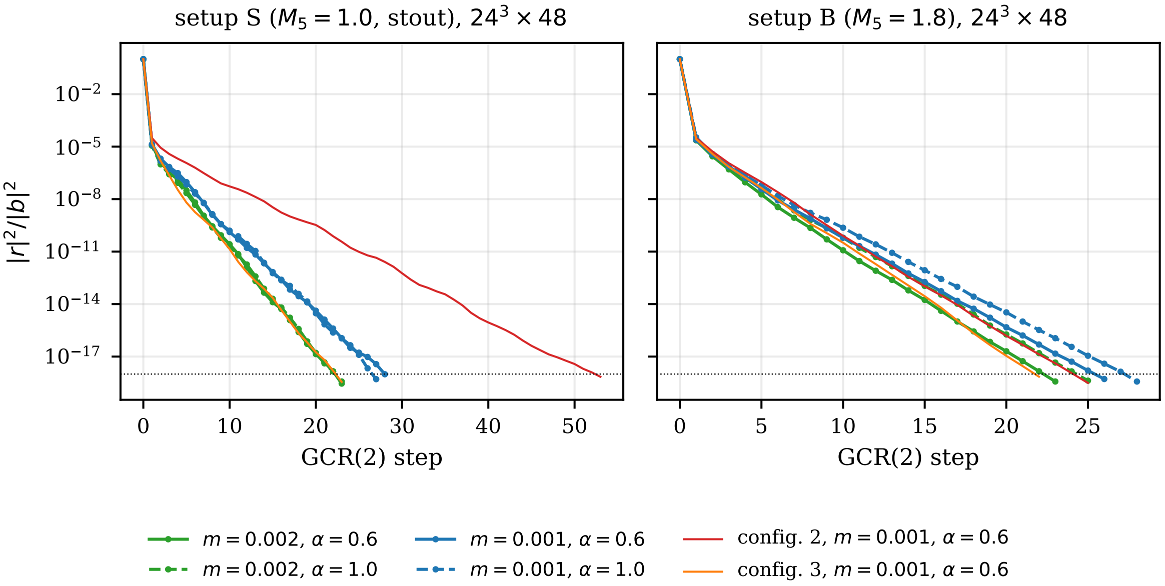

FIG. 8. GCR(2) residual histories on $24^3 \times 48$, $\tilde{A}_L$, the multigrid of Table 3: configuration 1 (blue, green) and configurations 2 and 3 at $m = 0.001$, $\alpha = 0.6$ (red, orange).

which matches the absence of slow setups without smearing. The converse does not hold: $16^4$ in setup S also reaches $-15.6$ and has no slow setups, and among the $12^4$ configurations the depth does not predict which exact cycles stagnate (configuration 2 reaches $-21.9$ and converges). Why smearing makes the coarse correction this large on the small lattice is open.

GCR(8) and 48 test vectors remove the slow setups on $12^4$ but do not fit $24^3 \times 48$ in 24 GB: each kept direction costs two fine vectors (1 GB there; GCR(4) peaks at 22.1 against 20.2 GB), and the setup with 48 test vectors runs out of device memory. Coarse solves for the quark factor to $10^{-4}$ cost no memory and leave 2 of 160 setups slow. On $24^3 \times 48$ they make each iteration 22–40% more expensive: configuration 2 drops to 33–43 iterations at about the same solve time, and the other three configurations take 9–34% longer. A slow setup shows itself in the first solve as a plateau of $|r|^2$; the setup can then be repeated with another seed on the same configuration, at the cost of one more setup. We have not automated this.

On $24^3 \times 48$ no run shows a plateau (Fig. 8); the first step reduces $|r|^2$ by about $10^{-5}$ in every run.

*What this establishes.* As usual for multigrid preconditioners of lattice Dirac operators, the convergence of the truncated flexible outer method is established numerically. GCR(2) with coarse solves stopped at $10^{-2}$ converged without plateaus in every run of this section on $16^4$ and $24^3 \times 48$, mostly at the rate the pseudospectral bound of Sec. V.D.3 gives for the full method; on $12^4$ with smearing about one setup in eight reaches a plateau (Table 19). No guarantee is available for a truncated method whose preconditioner varies from step to step. Accurate coarse solves do not make the solver faster. On $16^4$ the exact cycle takes 24–36 iterations on the four configurations, fewer than BiCGStab on the exact adjoint stopped at $10^{-2}$ (37–47 against 33–36 on configuration 1). The solver's cycle, with the regularized Pauli–Villars solve, takes 10–11. The cycle of the solver converged in every run of Table 20.

*In short.* The convergence is set by the spectrum and the pseudospectra, not by the field of values. The spectrum of $\tilde{A}_R\mathcal{M}$ stays at $\mathrm{Re}\,\lambda \geq 0.69$ with moderate condition numbers, and the pseudospectra give full GMRES a guaranteed rate of 0.5–0.6 per iteration, close to the 0.48–0.58 measured for GCR(2) on $24^3 \times 48$. The field of values reaches into the left half-plane because the coarse correction amplifies, and that loses the guarantee for truncated GCR. With the exact or the solver's cycle, GCR(2) stagnated only on $12^4$ with smearing. On configuration 1 the amplification is $1.9 \times 10^4$ on $12^4$ and $1.5 \times 10^4$ on $16^4$ (Table 15), and neither it nor the depth of the field of values alone decides which setups stagnate.

## VI. Summary

At $m = 0.001$ on four $24^3 \times 48$ quenched configurations, one GPU, the multigrid solve is 2.9–6.0 times faster than mixed-precision even-odd CGNE, and a point-source propagator takes 0.40–0.56 of its wall time, setup included. Its iterations grow by a factor of 1.6–4.1 from $m = 0.1$ to $m = 0.001$, those of CGNE by 18–27. The multigrid is the faster solver from $24^4$ on. BiCGStab and GCR(2) with the Chebyshev smoother on $\tilde{A}_L$ alone are slower than mixed-precision CGNE, so the gain comes from the coarse correction.

Each of the three reasons of Boyle and Yamaguchi is met by one choice: the even-odd system, a four-dimensional coarse lattice with the whole fifth dimension in each aggregate, and levels below the coarse lattice in place of deflation. The lift $\hat{t}$ is needed in setup S: without it GCR(2) stagnates or takes up to five times the iterations, although the correction is exact only for the right form. One setup serves every value of Neff's $\alpha$.

The field of values of the preconditioned operator reaches into the left half-plane, so the bound for GCR is lost. Its spectrum has $\mathrm{Re}\,\lambda \geq 0.69$ and its pseudospectra bound full GMRES with a fixed cycle. Truncated GCR(2) with accurate coarse solves stagnates on two of the four $12^4$ configurations in setup S, and with the solver's inexact cycle on about one setup in eight there. We saw no plateau on $16^4$ or $24^3 \times 48$, but truncated GCR has no guarantee.

Solving with $D^\dagger$, as HMC needs, uses the cycle of $D$ between two $\Gamma_5$, with no second setup. On $16^4$ it takes two to three iterations more than solving with $D$; on $12^4$ in setup S GCR(2) stalls on the residual right-hand side. Other constructions we tried were worse. HMC itself is not implemented here.

The evidence is four quenched configurations per volume up to $24^3 \times 48$ on one GPU, compared with CGNE and with single-grid Krylov solvers only. A comparison with HDCG and with the coarsening of $D_{\mathrm{PV}}^\dagger D$ in one code, larger volumes and HMC are the next steps (Sec. VII). Applying the coarse operators to the 12 columns of a point source together should raise the arithmetic intensity from about one flop per byte to about ten, as estimated for Wilson-clover multigrid [42]; we have not measured this.

*Code.* All results were obtained with RutticeQCD, `https://github.com/atsanroger/RutticeQCD`, a rewrite of Bridge++ [33, 34] in Rust whose GPU backend uses the CUDA kernels of the Wilson and Möbius operators of Bridge++. Where Bridge++ computes the same quantity, the results of RutticeQCD are compared with it number by number: the plaquette, the Wilson, clover and domain-wall operators applied to a vector, solver iteration counts, meson correlators, and single HMC trajectories. The domain-wall multigrid, which Bridge++ does not have, is checked against CGNE, whose propagators it reproduces (Sec. V.B.1).

## VII. Open problems

The following questions remain open. Those the measurements answer are discussed where they arise: the left form in Sec. III.A.3, the field of values in Secs. II.A and V.D.1–V.D.3.

1. **Parameters from the operator.** The smoother interval $[\lambda_{\mathrm{hi}}/\kappa_s, \lambda_{\mathrm{hi}}]$ uses a ratio chosen per setup, and a wrong one makes the solve stagnate (Sec. V.C.4); the setup works with singular values of $\tilde{A}_R$, which for a non-normal operator need not match its eigenvalues. How to set the interval from the operator, whether an ellipse with complex foci should replace the real interval, and how the required number of test vectors scales with the volume remain to be understood. Local coherence suggests that the number of test vectors does not grow with the volume, yet 16 suffice on $12^4$ and 24 do not on $24^3 \times 48$ (Sec. V.C.4). Part of this difference may come from the small physical box of $12^4$ ($\approx (1.2\,\mathrm{fm})^4$, with fewer low modes) and from the additional coarse levels used on $24^3 \times 48$ (Table 3); the low modes may also vary in the fifth dimension more than vectors not split in $s$ can follow (Eick and Mawhinney [52] find their fifth-dimensional profiles to differ between lattices);

a test vector split in $s$, as it is now split in chirality, would keep $s$ and $\hat{t}$ in the range of $\mathcal{P}$, and Lüscher's coherence test would measure what the coarse space captures.

2. **Convergence theory.** The pseudospectral bound rests on Lanczos estimates of $\sigma_{\min}$, which approach it from above, and on the eigenvalues farthest from the center found by Arnoldi (Sec. V.D.3). Obtaining a bound from an inclusion theorem, and tracking how the outliers of setup S and the pseudospectra develop beyond $16^4$, remain to be done; on $24^4$ the Arnoldi iteration does not fit in 24 GB of device memory. No theory covers truncated GCR with a preconditioner that varies from step to step. On $12^4$ in setup S, GCR(2) with accurate coarse solves can reach a residual orthogonal to its new direction and stay there (Sec. V.D.4), and on the residual right-hand side of HMC the solve with $D^\dagger$ needs GCR(8) (13–222 iterations) on $12^4$ in setup S (Sec. V.B.2). GCR(2) does not reach such a fixed point on $16^4$ and $24^4$, although $24^4$ has $Q = +4$; the solves with $D^\dagger$ were tested on $16^4$ but not on $24^4$. Why stout smearing gives the Galerkin Pauli–Villars factor singular values far below those of the fine operator (Sec. V.C.3) is not understood. A safeguard that changes the direction when $|\langle r|\tilde{A}\mathcal{M}r\rangle|/(\|r\|\,\|\tilde{A}\mathcal{M}r\|)$ falls below a threshold would remove such fixed points for one inner product per step; we have not implemented it. A quantitative explanation for why $N = c[Y, W]D^{-1}$ has a small square, beyond the heuristic of Sec. IV, is also lacking.

3. **Formulations and coarse spaces.** The comparisons here are with CGNE and with Krylov solvers on $\tilde{A}_L$ alone. How the two nine-point factors compare with the 81-point HDCG operator and with the 33-point coarsening of $D^\dagger_{\mathrm{PV}}D$ [43], in cost per iteration and in iterations, needs all of them in one code on the same lattices.

## Acknowledgments

I thank Issaku Kanamori and Hideo Matsufuru for their kind permission to port Bridge++ to Rust, without which this study would not have been possible; the structure of RutticeQCD follows Bridge++ closely, and its correctness rests on a number-by-number comparison with it. I am also grateful to my colleagues at my current company for teaching me the practical details of algebraic multigrid in computational fluid dynamics, which inspired the two-lift prolongation $\{s, \hat{t}\}$. Although computational fluid dynamics and lattice QCD lie far apart in physics, their numerical methods have much to learn from one another. Finally, I thank my friend Sakura Itatani for helping me find a reference on local coherence. I used Claude (Anthropic) to assist with the code; I take full responsibility for the content.

TABLE 5. The notation of this work and that of other references. Equation numbers are those of the references; "the same" means the same operator.

| | This work | Ref. [17] | Ref. [45] | Bridge++ code |
|---|---|---|---|---|
| Height | $M_5 > 0$ | $-M_5$, $M_5 = O(-1)$ (2.13) | $M_0$ | $M_0$, `domain_wall_height` |
| Fifth dim. | $s = 0, \dots, L_s - 1$ | $s = 1, \dots, L_s$ | $s = 1, \dots, N_s$ | $0, \dots, N_s - 1$, `extent_of_5th_dimension` |
| Wilson part | $W = (4 - M_5) - \frac{1}{2}K$ | $D^{\text{Wilson}}(M_5)$ (2.13) | $D_{\text{W}}(-M_0)$ | $\frac{1}{2\kappa}D_{\text{W}}$, $\kappa = 1/(8 - 2M_0)$ |
| $D_\pm$ | $D_+ = 1 + bW$,<br>$D_- = cW - 1$<br>(Eq. (2.3)) | $D_+ = bD^{\text{W}} + 1$,<br>$D_- = cD^{\text{W}} - 1$ (2.18) | the same (2) | — |
| $D$ | Eq. (2.7) | $D_{\text{DW}}(m)$ (2.20) | $D(m)$ (1) | `Fopr_Domainwall`, mode `D` |
| $C$ | $d_+ + d_- Y$, Eq. (2.14) | — | $\text{diag}(D_{ee}, D_{oo})$ (4) | $D_{ee}$, $D_{oo}$ of `_eo`;<br>`m_dp` $= \alpha d_+$, `m_dm` $= \alpha d_-$ |
| Neff's $\alpha$ | $DQ$, Eq. (2.21) | — (its $\alpha = \frac{b+c}{b-c}$) | — | `parameter_alpha`: $DQ$,<br>site part $CQ$ |
| $\Gamma_5$ | $\gamma_5 R$ | $\gamma_5 R D_-^{-1}$ (2.22) | — | — |
| 4d quark | $P_- \psi_0 + P_+ \psi_{L_s - 1}$ | $P_- \Psi_1$, $P_+ \Psi_{L_s}$ | — | the same |

# Appendices

## A. The notation of other references

Table 5 relates Eqs. (2.6)–(2.21) to Brower, Neff and Orginos [17], to Kanamori, Chen and Matsufuru [45] and to the code of Bridge++ [34] (`Fopr_Domainwall`, `Fopr_Domainwall_eo`, with Neff's $\alpha$ as `parameter_alpha`). Eq. (2.4) is the tridiagonal matrix of Ref. [17], Eq. (2.20), and of Ref. [45], Eq. (1), with $D_+ = 1 + bW$ on the diagonal, $D_- P_+$ below and $D_- P_-$ above it, $-mD_- P_\pm$ in the corners, and $s$ shifted by one. The operators $D$, $C$ and $D^{(\alpha)}$ are thus the same in all four; the differences are names and signs:

- $D_\pm$ are those of Refs. [17, 45], $D_- = cW - 1$. The hop $Y$ of Eq. (2.6) carries a minus sign, so $D = D_+ - D_- Y$ and $d_- = 1 - c(4 - M_5)$ is minus the site part of $D_-$.
- The height $M_5 > 0$ here is $M_0$ in Ref. [45] and in Bridge++ (`domain_wall_height`); Ref. [17] writes $M_5$ for the Wilson mass, $M_5^{\text{BNO}} = -M_5$, Eq. (2.13). The caption of Fig. 1 of Ref. [45] gives $M_0 = -1$ for the height 1 of its text.
- The $\alpha$ of Ref. [17], $(b + c)/(b - c)$, is the scale factor of the Möbius kernel (2 at $b = 1.5$, $c = 0.5$), not Neff's $\alpha$ of Eq. (2.21).
- $\Gamma_5 = \gamma_5 R D_-^{-1}$ in Ref. [17], Eq. (2.22); here $\Gamma_5 = \gamma_5 R$.

$D^{(\alpha)}$ here and the operator of Bridge++ with `parameter_alpha` were compared on a $12^4$ configuration with stout links, $L_s = 8$, $m = 0.01$: at $\alpha = 0.6$ and $\alpha = 1$ CG on the even-odd normal equation took the same number of iterations for every column of a point source, and the pseudoscalar correlators agree to $10^{-11}$. Bridge++ returns $\psi^{(\alpha)}$; $\psi = Q\psi^{(\alpha)}$.

## B. Spectra with other preconditioners

Figure 9 reproduces Fig. 3 of Ref. [45]. It shows the spectra that Sec. II.A compares with that of $A'$: the unpreconditioned $D$, the site-diagonal $DC^{-1}$, the Pauli–Villars form $D_{\text{PV}}^\dagger D$ of Ref. [41], and its block-diagonal approximations.

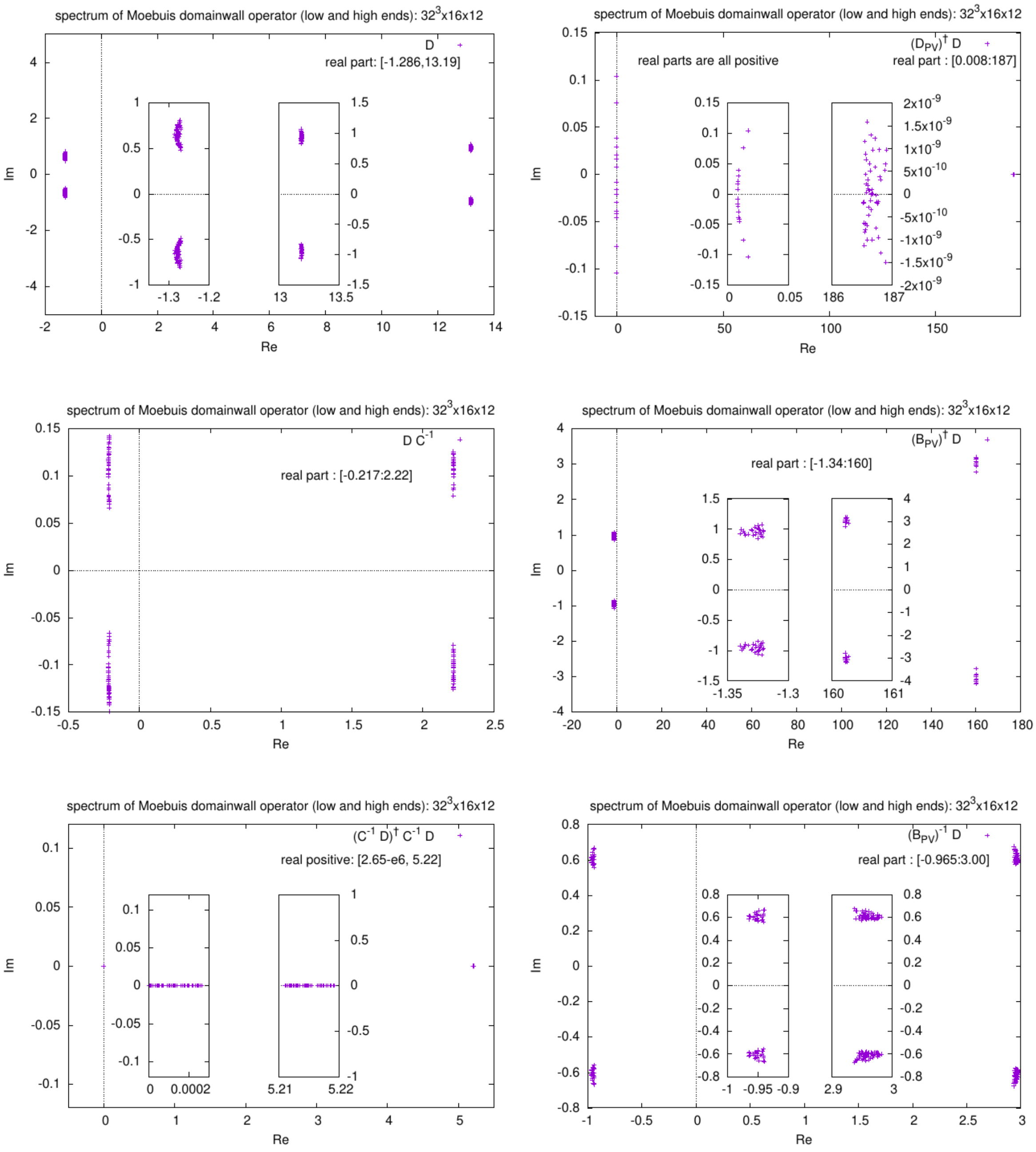

FIG. 9. Low and high ends of the spectrum of the Möbius operator on the configuration of Fig. 2, with preconditioners. Left column: $D$, $DC^{-1}$, $(C^{-1}D)^{\dagger}C^{-1}D$. Right column: $D_{\rm PV}^{\dagger}D$, $B_{\rm PV}^{\dagger}D$, $B_{\rm PV}^{-1}D$, where $B_{\rm PV}$ is the block-diagonal approximation of $D_{\rm PV}$ (inverted approximately in the last panel). From Ref. [45], Fig. 3.

TABLE 6. The gauge configurations: plaquette, $t_0/a^2$, and the topological charge at $t_0$. The logs and the generation histories are listed with the data.

| lattice | | plaquette | $t_0/a^2$ | $Q_{\text{clover}}$ | $Q_{\text{imp}}$ | $Q$ |
|---|---|---|---|---|---|---|
| $12^4$ | 1 | 0.594250 | 3.19 | +0.94 | +1.00 | +1 |
| | 2 | 0.593201 | 3.29 | +0.95 | +1.00 | +1 |
| | 3 | 0.594204 | 3.78 | +1.78 | +1.96 | +2 |
| | 4 | 0.594103 | 4.04 | +1.88 | +1.99 | +2 |
| $16^4$ | 1 | 0.593509 | 3.54 | +0.05 | +0.01 | 0 |
| | 2 | 0.593847 | 3.55 | +0.95 | +1.00 | +1 |
| | 3 | 0.594395 | 3.18 | +0.96 | +1.00 | +1 |
| | 4 | 0.593685 | 3.03 | +2.53 | +2.88 | +3 |
| $16^4$ dynamical | | 0.593809 | 6.90 | −0.003 | −0.0002 | 0 |
| $24^4$ | 1 | 0.593524 | 3.65 | +3.63 | +3.92 | +4 |
| | 2 | 0.593789 | 3.41 | +1.77 | +1.94 | +2 |
| | 3 | 0.593297 | 3.16 | −3.55 | −3.93 | −4 |
| | 4 | 0.593786 | 3.24 | −4.69 | −5.05 | −5 |
| $24^3 \times 48$ | 1 | 0.593548 | 3.20 | +4.47 | +4.78 | +5 |
| | 2 | 0.593630 | 3.18 | +7.83 | +8.42 | +8 |
| | 3 | 0.593689 | 3.19 | +3.46 | +3.53 | 3 or 4 |
| | 4 | 0.593395 | 3.23 | −1.57 | −1.87 | −2 |

### C. Gauge configurations

All quenched configurations use the plaquette gauge action at $\beta = 6.0$ and were generated by HMC from a cold start, 400 trajectories of length 0.5 (on $24^3 \times 48$, 400 from a cold start and then 200); these are configurations 1. Configurations 2–4 continue the same chains with trajectories of length 0.5 and are taken 100 trajectories apart, after 502, 602 and 702 trajectories (702, 802 and 902 on $24^3 \times 48$). The dynamical $16^4$ configuration was generated from the quenched one by HMC in setup S with mixed-precision CG: 60 trajectories at $m = 0.05$, then 40 at $m = 0.01$, of length 1 with Omelyan steps [20, 2]. Table 6 gives the plaquette, the scale $t_0$ ($t^2E = 0.3$, $E$ of the clover field strength) and the topological charge at $t_0$ after the Wilson flow (step 0.01), of the clover field strength and improved with rectangles ($c_{\text{plaq}} = 5/3$, $c_{\text{rect}} = -1/12$). On the larger lattices the two definitions differ by up to 1 at $t_0$.

### D. Krylov solvers

The multigrid uses four Krylov solvers, flexible GCR for the outer system, BiCGStab and CG for the coarse factors and FGMRES in the bootstrap, and the comparison uses CG as well. The algorithms below are those of RutticeQCD, which runs the same code on the CPU and on the device; $\epsilon$ is the tolerance on $|r|^2/|b|^2$.

*Flexible GCR($k$).* The outer solver of Sec. III.E is GCR [44] with a preconditioner $\mathcal{M}$ that may change from one step to the next, keeping the last $k$ directions (Algorithm 4). Each step applies $\mathcal{M}$ once, makes the new $q = \tilde{A}z$ orthogonal to the $q$ of the kept directions, and minimizes the residual along it; the step length $\langle q|r\rangle$ vanishes when $q$ is orthogonal to $r$, which is the fixed point of Sec. V.D.4. After $n_{\max}$ steps the true residual is recomputed and the solve restarts.

**Algorithm 4.** Flexible GCR($k$) for $\tilde{A}x = b$ with the preconditioner $\mathcal{M}$.

```
repeat
    $r \leftarrow b - \tilde{A}x$
    for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon|b|^2$ do
        $z \leftarrow \mathcal{M}r; \quad q \leftarrow \tilde{A}z$
        for each of the last $k$ pairs $(z_i, q_i)$ do
            $\gamma \leftarrow \langle q_i|q\rangle; \quad q \leftarrow q - \gamma q_i; \quad z \leftarrow z - \gamma z_i$
        end for
        $z \leftarrow z/|q|; \quad q \leftarrow q/|q|;$ keep $(z, q)$
        $\beta \leftarrow \langle q|r\rangle; \quad x \leftarrow x + \beta z; \quad r \leftarrow r - \beta q$
    end for
until $|b - \tilde{A}x|^2 < \epsilon|b|^2$
```

*FGMRES.* Each bootstrap round of Sec. III.C replaces a vector $y_j$ by a few steps of flexible GMRES [23] on $\tilde{A}_R x = y_j$ from $x = 0$, with the cycle as a right preconditioner (Algorithm 5). The basis is orthogonalized by classical Gram–Schmidt applied twice, since one pass loses orthogonality on the strongly non-normal $\tilde{A}_R\mathcal{M}$, and the least-squares problem of the Hessenberg matrix is updated by Givens rotations at every step.

**Algorithm 5.** FGMRES($n$) for $Ax = b$ with a right preconditioner $\mathcal{M}$.

```
repeat
    $r \leftarrow b - Ax; \quad \beta \leftarrow |r|; \quad v_0 \leftarrow r/\beta$
    for $k = 0, \ldots, n-1$, while the residual estimate is $\geq \epsilon^{1/2}|b|$ do
        $z_k \leftarrow \mathcal{M}v_k; \quad w \leftarrow Az_k$
        for two passes do
            for $i = 0, \ldots, k$ do $h_{ik} \leftarrow h_{ik} + \langle v_i|w\rangle; \quad w \leftarrow w - \langle v_i|w\rangle v_i$
            end for
        end for
        $h_{k+1,k} \leftarrow |w|; \quad v_{k+1} \leftarrow w/h_{k+1,k};$ update the Givens rotations and the residual estimate
    end for
    $c \leftarrow \arg\min_c |\beta e_0 - Hc|; \quad x \leftarrow x + \sum_i c_i z_i$
until $|b - Ax|^2 < \epsilon|b|^2$
```

*CGNE.* The reference solver of Sec. V.B.1 is CG on $\mathcal{K}^\dagger\mathcal{K}$ with $\mathcal{K} = C_e^{-1}S$, the even-odd operator of Bridge++, with Neff's $\alpha$ as for the multigrid, in double or in mixed precision.

For $D\psi = \eta$ the even-odd reduction of Sec. II.C gives $\mathcal{K}^{(\alpha)}y = C_e^{-1}\eta_e'$ with $\mathcal{K}^{(\alpha)} = Q_e^{-1}\mathcal{K}Q_e$, Eq. (2.27), and CG solves its normal equation $Ay = b$, $A = \mathcal{K}^{(\alpha)\dagger}\mathcal{K}^{(\alpha)}$, $b = \mathcal{K}^{(\alpha)\dagger}C_e^{-1}\eta_e'$; $\psi_o$ and $\psi = Q\psi^{(\alpha)}$ follow as for the multigrid. The stopping criterion is $|b - Ay|^2/|b|^2 < 10^{-18}$. Since the unknown is $y$ itself and $A$ has $\mathcal{K}^{(\alpha)}$ on the right, Bridge++ and Ref. [49] call this method CGNR; we keep the name CGNE that is common in lattice QCD.

*Mixed precision.* The mixed-precision solver runs the CG recurrence in single precision and corrects it by reliable updates [25, 26] in double precision (Algorithm 6): each time the single-precision $|\hat{r}|^2$ has fallen by a factor $\delta = 10^{-2}$ below its largest value since the last update, the single-precision solution is added to $y$ and the true residual $b - Ay$ is recomputed in double precision; CG then goes on with the same search direction and stops only on the double-precision residual. With $\delta = 0$ the algorithm is plain CG in double precision; in single precision it is the solver of the Pauli–Villars factor, Eq. (3.9), started from $x = 0$. On $24^3 \times 48$ the mixed-precision solver needs the iterations of double precision within $1\%$ except at $m \leq 0.002$ in setup S, where it needs up to $30\%$ more ($14\%$ on configuration 1), and takes 0.53–0.74 times the time (over the four configurations), with the same final residuals and 7.2 GB of device memory against 6.9 GB in double precision.

**Algorithm 6.** CGNE with reliable updates. Hats: single precision; $\hat{A}$ is $A$ on the single-precision links.

```
$r \leftarrow b - Ay$; if $|r|^2 < \epsilon|b|^2$ return $y$
$\hat{r} \leftarrow r$;   $\hat{p} \leftarrow \hat{r}$;   $\hat{x} \leftarrow 0$;   $\rho \leftarrow |\hat{r}|^2$;   $\rho_{\max} \leftarrow \rho$
loop
    $\alpha \leftarrow \rho / \langle \hat{p} | \hat{A}\hat{p} \rangle$;   $\hat{x} \leftarrow \hat{x} + \alpha\hat{p}$;   $\hat{r} \leftarrow \hat{r} - \alpha\hat{A}\hat{p}$;   $\rho' \leftarrow |\hat{r}|^2$
    if $\rho' < \delta\rho_{\max}$ or $\rho' < \epsilon|b|^2$ then                ▷ reliable update
        $y \leftarrow y + \hat{x}$;   $\hat{x} \leftarrow 0$;   $r \leftarrow b - Ay$
        if $|r|^2 < \epsilon|b|^2$ return $y$
        $\hat{r} \leftarrow r$;   $\rho' \leftarrow |\hat{r}|^2$;   $\rho_{\max} \leftarrow \rho'$
    else
        $\rho_{\max} \leftarrow \max(\rho_{\max}, \rho')$
    end if
    $\hat{p} \leftarrow \hat{r} + (\rho'/\rho)\,\hat{p}$;   $\rho \leftarrow \rho'$
end loop
```

*BiCGStab.* The coarse solves of the quark factor (Sec. III.B) use BiCGStab [22], Algorithm 7, with a right preconditioner $\mathcal{M}_A$. It bounds the angle between $t$ and $s$ from below as in Ref. [24], and after a breakdown or $n_{\max}$ steps it restarts from the true residual. The same algorithm, with $\mathcal{M}_A = 1$ and in double precision, is the Krylov solver on $\tilde{A}_L$ alone of Sec. V.C.1.

**Algorithm 7.** BiCGStab for $Ax = g$ with a right preconditioner $\mathcal{M}_A$; $\omega_c = 0.7$.

```
repeat
    $r \leftarrow g - Ax$;   $r_0 \leftarrow r$;   $p \leftarrow r$;   $\rho \leftarrow \langle r_0 | r \rangle$
    for at most $n_{\max}$ steps, while $|r|^2 \geq \epsilon |g|^2$ do
        $\hat{p} \leftarrow \mathcal{M}_A p$;   $v \leftarrow A\hat{p}$;   $\alpha \leftarrow \rho / \langle r_0 | v \rangle$;   $s \leftarrow r - \alpha v$
        $\hat{s} \leftarrow \mathcal{M}_A s$;   $t \leftarrow A\hat{s}$;   $\omega \leftarrow \langle t | s \rangle / |t|^2$
        $\gamma \leftarrow |\langle t | s \rangle| / (|t|\,|s|)$; if $\gamma < \omega_c$ then $\omega \leftarrow \omega\, \omega_c / \gamma$
        $x \leftarrow x + \alpha \hat{p} + \omega \hat{s}$;   $r \leftarrow s - \omega t$;   $\rho' \leftarrow \langle r_0 | r \rangle$
        $p \leftarrow r + (\rho'/\rho)(\alpha/\omega)\,(p - \omega v)$;   $\rho \leftarrow \rho'$
    end for
until $|g - Ax|^2 < \epsilon |g|^2$
```

*Two even-odd forms.* Boyle and Yamaguchi [40] used CG on $S^\dagger S$, which for Shamir fermions needed 3224 iterations against 3880 for the left-preconditioned form. Fig. 10 compares the two on the three quenched lattices at $m = 0.001$, in double precision with one random right-hand side, for $b - c = 1$ and five values of $c$; all other parameters are those of setups S and B.

The two operators differ by a site-local factor on the left, $S = C_e \mathcal{K}$, and with Neff's $\alpha$, $SQ_e = C_e Q_e \mathcal{K}^{(\alpha)}$. Their condition numbers therefore differ at most by the condition number of that factor. Since $C = d_+ + d_- Y$ with $\|Y\| = 1$ (Eq. (2.7)), the singular values of $C$ lie in $[d_+ - |d_-|,\, d_+ + |d_-|]$ and

$$\kappa(C) \leq \frac{d_+ + |d_-|}{d_+ - |d_-|}, \qquad d_+ = 1 + b(4 - M_5), \quad d_- = 1 - c(4 - M_5); \tag{D.1}$$

for $\alpha < 1$ the bound is $1/\alpha$ times larger, since $\kappa(Q) = 1/\alpha$. The measured ratios of the iterations stay within this range too, although the bound is on the condition numbers and not on the iteration counts.

Within the bound the order depends on $c$. In setup B the ratio of the iterations grows smoothly with $c$, from 0.81–0.83 for Shamir fermions to 1.27–1.31 at $(b, c) = (2, 1)$, and crosses 1 near the $(b, c) = (1.5, 0.5)$ used in the main text. For Shamir fermions $S^\dagger S$ is the faster form by 1.20–1.23, close to the $3880/3224 = 1.20$ of Ref. [40]. In setup S the ratio is above 1 except at $c = 0.25$, where $S^\dagger S$ needs 3–5% fewer iterations; we have no explanation for this dip. The three lattices agree to within 0.04 in every case, so the order is a property of the operator and its parameters and does not depend on the configuration.

Neff's $\alpha$ favors $\mathcal{K}^\dagger \mathcal{K}$. With $\alpha = 0.6$, $S^\dagger S$ needs 46–58% more iterations in setup S and 14–32% more in setup B, for every $c$ and on all three lattices, Shamir fermions included. In $\mathcal{K}^{(\alpha)}$ the factor $C_e Q_e$ is divided out on the left, whereas in $SQ_e$ the rescaling acts on the right only. The singular values of the two operators would show whether this is the cause. For the setups of this work $\mathcal{K}^\dagger \mathcal{K}$ is the faster form, and the multigrid is compared with it.

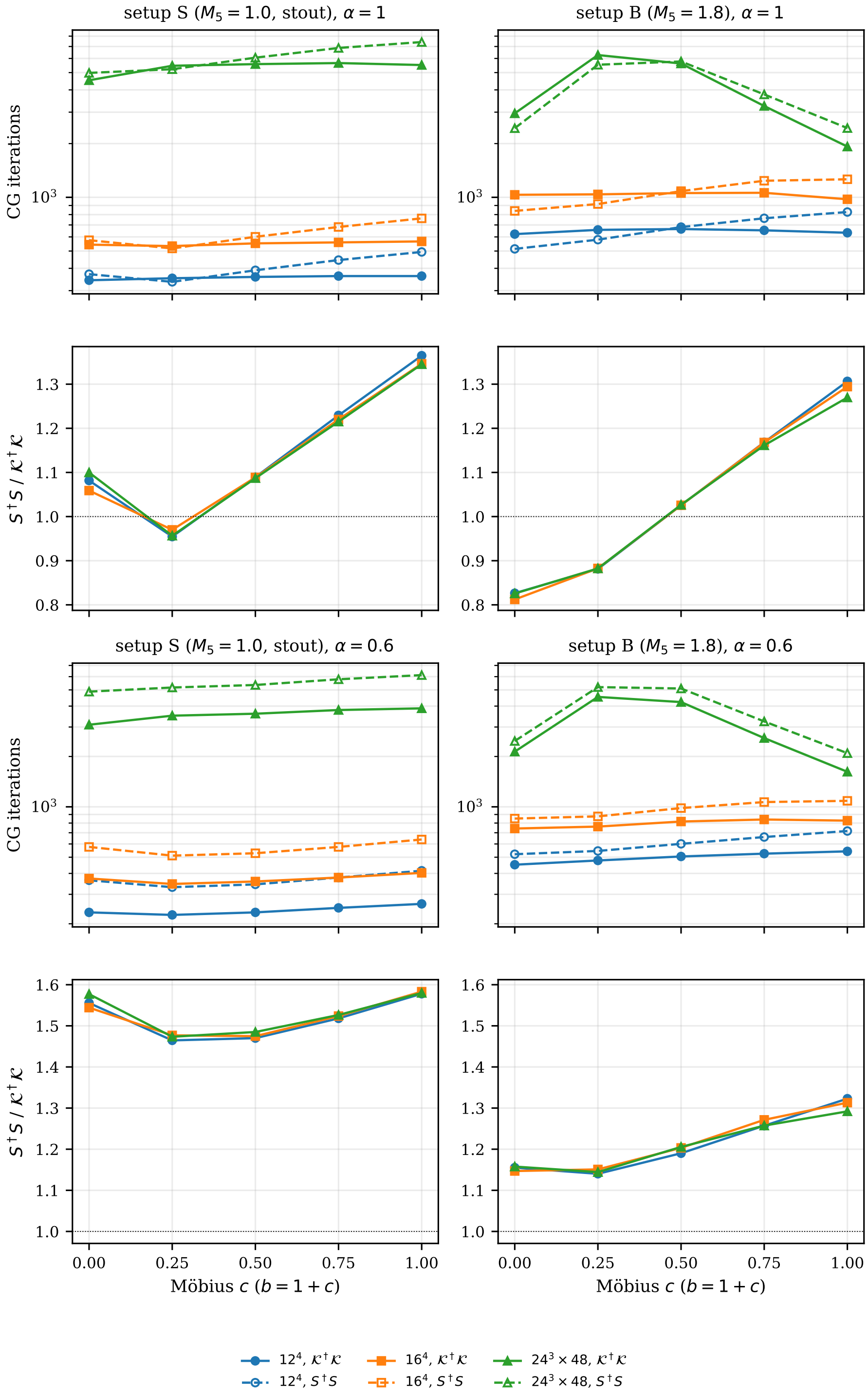

FIG. 10. CG iterations on $\mathcal{K}^\dagger\mathcal{K}$ (filled, solid) and on $S^\dagger S$ (open, dashed), with $\mathcal{K} = 1 - C_e^{-1} D_{eo} C_o^{-1} D_{oe}$ (the even-odd operator of Bridge++) and $S = C_e - D_{eo} C_o^{-1} D_{oe}$ (the Schur complement of Grid's default `SchurDiagMooeeOperator`, there on the odd sites), against the Möbius $c$, $b = 1 + c$, and below each their ratio; above 1 the left-preconditioned form is faster. Upper four panels $\alpha = 1$, lower four $\alpha = 0.6$.

TABLE 7. Mass scan on $24^3 \times 48$, means over the four configurations, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$ on the even-odd system: even-odd CGNE in double and in mixed precision against the multigrid (GCR(2) on $\tilde{A}_L$), $\alpha = 0.6$ for both and $\alpha = 1$ for the multigrid; iterations, solve time in seconds, coarse-operator applications, and $N_{\mathrm{rhs}}$, the number of right-hand sides at which the multigrid setup (194–237 s) is paid back, $t_{\mathrm{setup}}/(t_{\mathrm{CGNE}} - t_{\mathrm{MG}})$, against the double- and the mixed-precision CGNE. The mixed-precision CGNE needs the iterations of the double-precision one within $1\%$, except at $m \leq 0.002$ (S), where it needs up to $30\%$ more. Final $|D\psi - \eta|^2/|\eta|^2$: $9 \times 10^{-18}$–$8 \times 10^{-15}$ (CGNE, both), $2 \times 10^{-19}$–$3 \times 10^{-18}$ (MG). Device memory peak: CGNE 6.9 GB (double) and 7.2 GB (mixed), MG 20.2 GB ($\alpha = 0.6$). Table 8 gives the four configurations at $m = 0.001$.

| | CGNE | | | MG, $\alpha = 0.6$ | | | MG, $\alpha = 1$ | | $N_{\mathrm{rhs}}$ | |
|---|---|---|---|---|---|---|---|---|---|---|
| $m$ | it | s | s, mixed | it | s | coarse | it | s | double | mixed |
| setup S ($M_5 = 1.0$, stout) | | | | | | | | | | |
| 0.1 | 182 | 5.0 | 2.9 | 14 | 6.1 | 1336 | 18 | 7.4 | — | — |
| 0.05 | 303 | 8.3 | 4.7 | 13 | 5.2 | 808 | 14 | 5.7 | 63 | — |
| 0.02 | 658 | 17.9 | 9.7 | 14 | 5.6 | 874 | 14 | 5.8 | 16 | 48 |
| 0.01 | 1174 | 31.9 | 17.1 | 16 | 6.7 | 1168 | 16 | 6.7 | 7.9 | 19 |
| 0.005 | 2079 | 56.5 | 30.1 | 20 | 8.3 | 1589 | 20 | 8.6 | 4.3 | 9.4 |
| 0.002 | 3424 | 92.9 | 50.4 | 26 | 11.7 | 2684 | 26 | 12.2 | 2.5 | 5.3 |
| 0.001 | 3668 | 99.5 | 61.3 | 32 | 15.4 | 4012 | 32 | 15.8 | 2.5 | 4.6 |
| setup B ($M_5 = 1.8$) | | | | | | | | | | |
| 0.1 | 234 | 6.4 | 3.7 | 7 | 3.2 | 204 | 9 | 4.1 | 65 | 440 |
| 0.05 | 405 | 11.1 | 6.1 | 10 | 4.6 | 310 | 12 | 5.5 | 32 | 140 |
| 0.02 | 866 | 23.6 | 12.7 | 13 | 6.0 | 436 | 15 | 6.9 | 12 | 31 |
| 0.01 | 1481 | 40.2 | 21.5 | 16 | 7.2 | 575 | 17 | 7.9 | 6.3 | 15 |
| 0.005 | 2354 | 63.9 | 34.0 | 18 | 8.4 | 739 | 20 | 9.5 | 3.8 | 8.3 |
| 0.002 | 3628 | 98.5 | 52.3 | 22 | 10.3 | 1000 | 24 | 11.6 | 2.4 | 5.1 |
| 0.001 | 4456 | 120.9 | 64.2 | 24 | 11.5 | 1172 | 26 | 12.6 | 1.9 | 4.0 |

### E. Tables of the numerical results

The tables that support Sec. V, in the order in which it cites them.

TABLE 8. The four $24^3 \times 48$ configurations at $m = 0.001$, $\alpha = 0.6$ (Table 7): iterations and solve seconds, and the solve time of each CGNE over that of the multigrid.

| | | CGNE, double | | CGNE, mixed | | MG | | speed-up | |
|---|---|---|---|---|---|---|---|---|---|
| setup | config. | it | s | it | s | it | s | double | mixed |
| S | 1 | 3602 | 98.5 | 4090 | 59.9 | 28 | 13.2 | 7.5 | 4.5 |
| S | 2 | 3785 | 102.5 | 4936 | 76.1 | 53 | 26.0 | 3.9 | 2.9 |
| S | 3 | 3584 | 96.9 | 3622 | 51.9 | 23 | 10.3 | 9.4 | 5.0 |
| S | 4 | 3702 | 100.1 | 3990 | 57.2 | 25 | 12.3 | 8.1 | 4.7 |
| B | 1 | 4228 | 115.6 | 4232 | 62.0 | 26 | 12.9 | 9.0 | 4.8 |
| B | 2 | 4955 | 134.1 | 4959 | 70.9 | 25 | 11.8 | 11.4 | 6.0 |
| B | 3 | 4269 | 115.5 | 4276 | 61.1 | 22 | 10.3 | 11.2 | 5.9 |
| B | 4 | 4371 | 118.3 | 4377 | 62.7 | 23 | 10.9 | 10.9 | 5.8 |

TABLE 9. Iterations and solve seconds at $m = 0.001$ against the volume, one random right-hand side: ranges over the four configurations; multigrid: plus its mean setup time. BiCGStab: configuration 1. The 209 iterations on $12^4$ (S) are a slow setup of Sec. V.D.4.

| | | | $12^4$ | $16^4$ | $24^4$ | $24^3 \times 48$ |
|---|---|---|---|---|---|---|
| S | CGNE, mixed | it | 234–289 | 358–781 | 1715–2657 | 3622–4936 |
| | | s | 0.2 | 0.6–1.2 | 12.5–19.4 | 51.9–76.1 |
| | BiCGStab on $\tilde{A}_L$ | it | 235 | 398 | 1905 | 3247 |
| | | s | 0.4 | 2.0 | 49.9 | 169.5 |
| | multigrid | it | 12–209 | 12–14 | 15–20 | 23–53 |
| | | s | 0.5–7.8 (+7) | 0.6–0.8 (+18) | 2.9–4.4 (+96) | 10.3–26.0 (+209) |
| B | CGNE, mixed | it | 505–584 | 818–1249 | 3388–4116 | 4232–4959 |
| | | s | 0.3–0.4 | 1.3–1.9 | 24.6–29.8 | 61.1–70.9 |
| | BiCGStab on $\tilde{A}_L$ | it | 368 | 663 | 1909 | 2085 |
| | | s | 0.7 | 3.4 | 50.0 | 108.8 |
| | multigrid | it | 13–14 | 14–16 | 22–23 | 22–26 |
| | | s | 0.4–0.5 (+6) | 0.8–0.9 (+18) | 5.2–5.5 (+102) | 10.3–12.9 (+213) |

TABLE 10. GCR iterations of a solve with $D^\dagger$ to $|r|^2/|b|^2 < 10^{-10}$ from one setup of $D$ (32 test vectors, blocks $4^4$). "rand": random right-hand side; "res": the residual after one sweep of $D^\dagger D$. Medians over the four configurations, in brackets the number that do not converge in 300 iterations ("stall": three or four; with two, the range of the other two). One inner cycle per step. For comparison, the solve with $D$ takes 7–9 ($16^4$) and 7–12 ($12^4$) iterations on configuration 1.

| | $16^4$, $m = 0.01$ | | | | $12^4$, $m = 0.001$ | | | |
|---|---|---|---|---|---|---|---|---|
| | S | | B | | S | | B | |
| method | rand | res | rand | res | rand | res | rand | res |
| dual cycle, GCR(2) | 16 | stall | 9 | 13 | stall | stall | 13 | 17 |
| dual cycle, GCR(8) | 10 | 20 [1] | 9 | 12 | stall | stall | 11 | 16 |
| $\Gamma_5\mathcal{M}\Gamma_5$, GCR(2) | 9 | 10 | 11 | 11 | 12–13 [2] | stall | 12 | 14 |
| $\Gamma_5\mathcal{M}\Gamma_5$, GCR(8) | 9 | 10 | 10 | 11 | 11 | 20 | 12 | 12 |

TABLE 11. GCR(2) iterations on $\tilde{A}_L$ with $\mathcal{P}$ made of different lifts of the vectors $y$, $m = 0.001$, one random right-hand side; the other parameters as in Table 3; setup seconds on $24^3 \times 48$. $12^4$ and $16^4$: medians over the four configurations, in brackets the number that do not converge in 2000 iterations ("stag.": three or four); $24^3 \times 48$: configuration 1.

| | | | $12^4$ | | $16^4$ | | $24^3 \times 48$ | | setup (s) |
|---|---|---|---|---|---|---|---|---|---|
| lifts | $y$ | columns | S | B | S | B | S | B | S, B |
| $\{s, \hat{t}\}$ | 16 | 32 | 14 | 14 | 13 | 15 | 28 | 26 | 196, 211 |
| $\{s, Ds\}$ | 16 | 32 | stag. | 30 | stag. | 22 | 148 | 30 | 193, 207 |
| $\{s\}$ | 16 | 16 | stag. | 31 | stag. | 22 | stag. | 32 | 143, 153 |
| $\{s\}$ | 32 | 32 | 90 [1] | 32 | 27 | 15 | 28 | 13 | 346, 337 |

TABLE 12. GCR(2) iterations on $\tilde{A}_L$ for seven setups (configuration, seed) of setup S on $12^4$, $m = 0.001$: without the shift (BiCGStab on the exact adjoint), with $\tau = 10^{-3}$, and the range over $\tau = 10^{-4}$, $3 \times 10^{-4}$, $3 \times 10^{-3}$, $10^{-2}$, $3 \times 10^{-2}$; otherwise Table 3. n.c.: not converged in 2000 iterations. *One run, $\tau = 10^{-2}$: 63.

| | no shift | $\tau = 10^{-3}$ | other $\tau$ |
|---|---|---|---|
| 1, seed 2 | n.c. | 15 | 11–13 |
| 1, seed 5 | 70 | 14 | 13–19 |
| 1, seed 6 | 270 | 14 | 13–14 |
| 2, seed 10 | 106 | 12 | 12–14 |
| 3, seed 3 | n.c. | 13 | 12–15 |
| 3, seed 10 | 204 | 13 | 11–14* |
| 4, seed 21 | n.c. | 18 | 14–16 |

TABLE 13. Smallest singular value of the coarse Pauli–Villars factor $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} = \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P}$ against that of the fine $D_{\mathrm{PV}}$ (Lanczos on $X^\dagger X$), setup S, $m = 0.001$, seed 21. Stout: the smearing of setup S; none: the same setup without smearing.

| | | configuration 1 | | configuration 4 | |
|---|---|---|---|---|---|
| | smearing | $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$ | $D_{\mathrm{PV}}$ | $\mathcal{D}_{\mathrm{c}}^{\mathrm{PV}}$ | $D_{\mathrm{PV}}$ |
| $12^4$ | stout | $1.28 \times 10^{-2}$ | 0.532 | $3.32 \times 10^{-3}$ | 0.261 |
| $12^4$ | none | 0.259 | 0.262 | 0.233 | 0.242 |
| $16^4$ | stout | $1.42 \times 10^{-2}$ | 0.416 | $2.76 \times 10^{-2}$ | 0.236 |

TABLE 14. The changes of Fig. 6 that move the cost, with timings: iterations / setup seconds / solve seconds, medians over the four configurations as in Fig. 6. $12^4$, $m = 0.001$, $\tilde{A}_L$, GCR(2), one random right-hand side.

| change | setup S | setup B |
|---|---|---|
| base | 14 / 7 / 0.5 | 14 / 6.5 / 0.5 |
| aggregates $2^4$ | 8 / 19 / 1.4 | 9 / 12 / 1.2 |
| aggregates $3^4$ | 12 / 8 / 0.5 | 11 / 7 / 0.4 |
| aggregates $6^4$ | 38 [1] / 7 / 1.1 | 20 / 7 / 0.5 |
| test vectors 16 | 26 [1] / 4 / 0.6 | 19 / 4 / 0.5 |
| test vectors 24 | 18 / 5.5 / 0.5 | 16 / 5.5 / 0.4 |
| test vectors 48 | 10 / 16 / 0.7 | 11 / 12 / 0.8 |
| smoother $\nu = 4$ | 30 / 8 / 1.2 | stag. |
| smoother $\nu = 6$ | 16 / 7 / 0.6 | 81 [1] / 6 / 2.6 |
| smoother $\nu = 12$ | 12 / 7.5 / 0.6 | (base) |
| smoother $\nu = 16$ | 9 / 8 / 0.5 | 13 / 7 / 0.5 |
| bootstrap rounds 1 | 32 [1] / 4 / 1.1 | 40 [1] / 4 / 0.9 |
| bootstrap rounds 3 | 18 / 9.5 / 0.7 | 13 / 9 / 0.5 |

TABLE 15. Real part of the field of values $W(X)$ and $\|I - X\|_2$ for $X = \tilde{A}_R\mathcal{S}$, $\tilde{A}_R E$, $\tilde{A}_R\mathcal{M}$ (exact coarse solves). $m = 0.05$ on $4^3 \times 8$ ($2^4$ aggregates, 8 filtered low modes), $m = 0.001$ otherwise, with the multigrid of the mass scan (16 low modes, 32 test vectors, $4^4$, Chebyshev of Table 3) but no levels below the coarse lattice. Rows "4 configs." and "no smearing": ranges over the four $12^4$ configurations, $M_5 = 1$; other rows configuration 1. On $24^4$ only $W(\tilde{A}_R)$ is given (Krylov dimensions 10 and 12 agree to six digits); there the field of values of $\tilde{A}_R\mathcal{M}$ did not settle (the two dimensions differ by $2\%$) and the eigensolvers did not fit in 24 GB of device memory.

|  | $\mathrm{Re}\, W(\tilde{A}_R)$ | $\|I - \tilde{A}_R\mathcal{S}\|$ | $\|I - \tilde{A}_R E\|$ | $\mathrm{Re}\, W(\tilde{A}_R\mathcal{M})$ | $\|I - \tilde{A}_R\mathcal{M}\|$ |
|---|---|---|---|---|---|
| $4^3 \times 8$, $M_5 = 1$ | $[-5.0 \times 10^{-3}, 2.26]$ |  | 18.5 | $[0.772, 1.222]$ |  |
| $12^4$ S | $[-4.59 \times 10^{-3}, 1.98]$ | 1.018 | $1.94 \times 10^4$ | $[-23.9, 30.8]$ | 54.8 |
| $12^4$ S, 4 configs. | $\geq -0.013$ | 1.02–1.13 | $5 \times 10^3$–$3 \times 10^5$ | $\geq -324$ | 15–722 |
| $12^4$ S, no smearing | $\geq -0.011$ | 1.07–1.08 | 64–84 | $[0.23, 2.62]$ | 1.5–2.0 |
| $12^4$ B | $[-5.02 \times 10^{-2}, 4.46]$ | 1.112 | $2.86 \times 10^3$ | $[-1.49, 4.17]$ | 5.64 |
| $16^4$ S |  |  | $1.54 \times 10^4$ | $[-15.6, 27.1]$ |  |
| $16^4$ B |  |  | $2.5 \times 10^3$ | $[-1.42, 4.27]$ |  |
| $24^4$ S | $[-1.31 \times 10^{-2}, 1.98]$ |  |  |  |  |
| $24^4$ B | $[-5.08 \times 10^{-2}, 4.48]$ |  |  |  |  |

TABLE 16. Eigenvalues of $\tilde{A}\mathcal{M}$ (exact coarse solves) by Krylov–Schur: the smallest real part, the condition numbers $\kappa = 1/|y^\dagger x|$ of the six eigenvalues of smallest real part (left eigenvectors from $\mathcal{M}^\dagger \tilde{A}^\dagger$), the eigenvalues farthest from 1 with their $\kappa$, and the radius about 1 of the disc that holds all the others. $m = 0.001$. On $16^4$ the smallest real part and $\kappa$ are ranges over the four configurations, the outliers and the radius are those of configuration 1.

| | $\min \operatorname{Re} \lambda$ | $\kappa$ (6 lowest) | outliers ($\kappa$) | others: $\lvert\lambda - 1\rvert \leq$ |
|---|---|---|---|---|
| $4^3 \times 8, m = 0.05$, R | 0.857 | 2–4 | | |
| $4^3 \times 8, m = 0.05$, L | 0.854 | 2.8 | | 0.156 (Ritz) |
| $12^4$ S, R | 0.868 | 6.3–14.5 | $5.86 + 0.86i$ (11.2) | 0.434 |
| $12^4$ S, L | 0.861 | 7.2–26.2 | $5.40 + 0.89i$ (13.9) | 0.455 |
| $12^4$ B, R | 0.831 | 9.4–20.3 | | 0.753 |
| $12^4$ B, L | 0.830 | | | 0.747 |
| $16^4$ S, R | 0.700–0.794 | 3.6–22.8 | $10.63 + 0.19i$ (4.38) | 0.474 |
| | | | $3.13 + 0.02i$ (4.74) | |
| $16^4$ B, R | 0.698–0.783 | 9.5–42.9 | | 0.840 |

TABLE 17. $\min_\Gamma \sigma_{\min}(z - \tilde{A}_R \mathcal{M})$ on circles $|z - c| = r$ (16 points each, or the number given), and the bound (5.1) or (5.2): the factor per iteration $r/c$, the constant, and the iterations it guarantees for $\|r_k\|/\|r_0\| < 10^{-9}$. For the dense circles $\varepsilon$ is the lower bound $\min_i \sigma_{\min}(z_i) - h/2$ on the whole circle, $h$ the spacing of the points ($\sigma_{\min}$ is 1-Lipschitz in $z$); otherwise the smallest sampled value. Lanczos to $10^{-3}$ relative in $\sigma^2$.

| | $c$ | $r$ | $\min \sigma_{\min}$ | $r/c$ | constant | $k$ for $10^{-9}$ |
|---|---|---|---|---|---|---|
| $12^4$ S | 1 | 0.6 | 0.0285 | 0.6 | 220 | 53 |
| $12^4$ S, 160 points | 1 | 0.6 | 0.0281, $\geq 0.0164$ | 0.6 | 380 | 54 |
| $12^4$ S | 1 | 0.5 | 0.0107 | 0.5 | 480 | 40 |
| $12^4$ B | 1.4 | 0.93 | 0.0985 | 0.66 | 9.4 | 57 |
| $12^4$ B, 96 points | 1.4 | 0.93 | 0.0980, $\geq 0.0676$ | 0.66 | 13.8 | 58 |
| $12^4$ B | 1.3 | 0.87 | 0.0860 | 0.67 | 10.1 | 58 |
| $16^4$ S | 1 | 0.6 | 0.0262 | 0.6 | 190 | 53 |
| $16^4$ S | 1 | 0.55 | 0.0135 | 0.55 | 340 | 47 |
| $16^4$ B | 1.4 | 1.05 | 0.1295 | 0.75 | 8.1 | 80 |

TABLE 18. GCR($k$) iterations to $|r|^2/|b|^2 < 10^{-18}$, $12^4$, $m = 0.001$, the multigrid of Table 3 (no levels below the coarse lattice), one random right-hand side: medians over the four configurations, in brackets the number that do not converge in 1000 iterations; the point source: range over its 12 columns and the configurations. *Configuration 4 (the slow setup of Table 19): 97–521, one column not converged.

| | coarse solves | $k = 2$ | 8 | 16 | 32 | 64 |
|---|---|---|---|---|---|---|
| S, $\tilde{A}_L$ | $10^{-2}$ | 14 | | | 12 | |
| S, $\tilde{A}_R$ | $10^{-2}$ | 12 | | | 11 | |
| B, $\tilde{A}_L$ | $10^{-2}$ | 14 | | | 13 | |
| B, $\tilde{A}_R$ | $10^{-2}$ | 15 | | | 14 | |
| S, $\tilde{A}_L$, point source | $10^{-2}$ | 12–19* | | | 12–15 | |
| S, $\tilde{A}_L$ | $10^{-8}$ | no conv. | 24 [1] | 17 | 17 | 17 |
| S, $\tilde{A}_L$ | $10^{-4}$ | 18 [1] | | | 16 | |

TABLE 19. Slow setups: setups with more than 30 iterations or no convergence in 2000 / setups (seeds per configuration times configurations), and the range of the others; $m = 0.001$, $L_s = 8$, $\tilde{A}_L$, one random right-hand side to $|r|^2/|b|^2 < 10^{-18}$. $12^4$: configurations 1–4; $16^4$: 1–3 and the dynamical one; $24^3 \times 48$: 3 and 4. Last column: peak device memory of the variant on $24^3 \times 48$ (setup S, configuration 2, seed 21).

| lattice | setup | variant | seeds per conf. | slow | others | memory |
|---|---|---|---|---|---|---|
| $12^4$ | S | GCR(2) | 1–40 | 19/160 | 11–30 | 20.2 GB |
|  | S | GCR(4) | 1–10 | 3/40 | 12–20 | 22.1 GB |
|  | S | GCR(8) | 1–10 | 0/40 | 11–19 | does not fit |
|  | S | GCR(16) | 1–10 | 0/40 | 11–14 | does not fit |
|  | S | 48 test vectors | 1–10 | 0/40 | 9–15 | does not fit |
|  | S | quark coarse $10^{-4}$ | 1–40 | 2/160 | 11–30 | 20.2 GB |
|  | S, no smearing | GCR(2) | 1–10 | 0/40 | 13–15 |  |
|  | B | GCR(2) | 1–20 | 0/80 | 13–14 |  |
| $16^4$ | S | GCR(2) | 1–20 | 0/60 | 12–15 |  |
|  | S, dynamical | GCR(2) | 1–20 | 0/20 | 11 |  |
| $24^3 \times 48$ | S | GCR(2) | 1–5 | 0/10 | 18–28 |  |

TABLE 20. GCR(2) on $\tilde{A}_L$ in setup S, $m = 0.001$, $\alpha = 0.6$, to $|r|^2/|b|^2 < 10^{-16}$ (at most 500 iterations, 300 on $24^4$): converged runs / runs, with their iterations. “exact”: coarse solves to $10^{-20}$ in double precision, BiCGStab on the exact Pauli–Villars adjoint; “fixed”: the exact cycle plus $\delta\,\Gamma_5\mathcal{M}r$ or plus a fixed multiple of the smoother polynomial, $\delta$ the relative change that coarse solves to $10^{-2}$ make (0.04–0.23); “random”: the exact cycle plus a random vector of relative size $\delta = 0.02$–$0.5$, drawn anew at each application; “solver”: the cycle of the solver (MINRES, $10^{-2}$), whose correction did not fit next to the exact one on $24^4$ (OOM). $Q$: the topological charge. Aggregates $4^4$ unless stated; configurations 2–4: three right-hand sides.

| lattice, $Q$ | exact | fixed $\Gamma_5\mathcal{M}$ | fixed $p(\tilde{A}_L)$ | random | solver |
|---|---|---|---|---|---|
| $12^4$ config. 1, $+1$ | 0/5 | 0/5 | 0/5 | 75/75 (34–120) | 5/5 (12–14) |
| $12^4$ config. 2, $+1$ | 3/3 (21–24) |  |  |  | 3/3 (12–13) |
| $12^4$ config. 3, $+2$ | 3/3 (18–20) |  |  |  | 3/3 (10) |
| $12^4$ config. 4, $+2$ | 0/3 |  |  |  | 3/3 (84–94) |
| $12^4$, $3^4$ aggregates | 0/3 |  |  |  | 3/3 (11–12) |
| $12^4$, $2^4$ aggregates | 2/3 (20, 23) |  |  |  | 3/3 (7) |
| $16^4$ config. 1, 0 | 5/5 (33–36) | 5/5 | 5/5 | 75/75 (31–114) | 5/5 (12) |
| $16^4$ config. 2, $+1$ | 3/3 (24–25) |  |  |  | 3/3 (12–13) |
| $16^4$ config. 3, $+1$ | 3/3 (29–31) |  |  |  | 3/3 (11) |
| $16^4$ config. 4, $+3$ | 3/3 (25–29) |  |  |  | 3/3 (12) |
| $16^4$ dynamical, 0 | 5/5 (13) | 5/5 | 5/5 | 20/20 (13–20) | 5/5 (10) |
| $24^4$, $+4$ | 2/2 (78, 85) | 0/2 | 2/2 (41, 42) | 4/4 (84–205) | OOM |

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